

Differentiable Discrete Symplectic Cosmology: Forward Simulation and Inverse Reconstruction of CMB-like Fluctuations

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Abstract

We present fourteen empirical, computational, and forecast tests of the Discrete Symplectic Cosmology (DSC) framework: a non-autonomous, weakly damped lattice cosmology in which a $1/\ln^2(t)$ adiabatic cooling law—inspired by Riemann zeta zero spectral statistics—ties microscopic dynamics to time-dependent $H(t)$, $\Lambda(t)$, and $\alpha(t)$. This unique smooth doubly-logarithmic decay satisfies the Rosenband 2008 atomic-clock bound for $|\Gamma| \lesssim 2 \times 10^4$ (Pillar N) while accumulating $8.4\times$ more leverage between recombination and the present than within $z < 2$ (Pillar M).

Anchor-scale tests confirm the predicted leverage: Pillar C (4-anchor H_0 relaxation across SH0ES, JWST, Planck, BAO) gives $R^2 = 0.91$; Pillar A (Murphy 2003 Keck/HIRES, 128 absorbers) gives $\Gamma = +5.56 \pm 0.99$ (5.6σ under fit-error weighting, 2.8σ under Murphy’s σ_{rand} budget); Pillar K cross-validates on King 2012 Keck-only ($+6.5\sigma$, union $+8.6\sigma$). Dense-scale tests confirm the predicted *unresolvability*: on cosmic chronometers, Pantheon+, DESI DR2, and the 1623-constraint joint set (Pillars F–J), DSC is degenerate with constant H_0 over $z < 2$ — a fixed property of $1/\ln^2$, not a tunable feature. The King 2012 VLT-only subset gives Γ of opposite sign (-3.7σ), reducing the Keck+VLT union to $+3.0\sigma$, an instrumental systematic. A Störmer–Verlet lattice surrogate of CMB anisotropies yields $\chi^2/\text{dof} \approx 80$ against Planck-2018; we retract any quantitative C_ℓ claim. Pillar O: five JWST chronometers at $z \in [2, 5]$ distinguish DSC from ΛCDM at $\gtrsim 11\sigma$.

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1. Introduction

The temperature anisotropies of the cosmic microwave background (CMB), as mapped by the Planck satellite [1, 2], encode a remarkably simple statistical portrait of the early universe: the fluctuations are nearly Gaussian [3], their angular power spectrum exhibits a series of acoustic peaks at multipoles $\ell \sim 220, 540, 810, \dots$ [4], and the comoving sound horizon at recombination is finite [5]. In the standard Λ CDM paradigm, these features arise from quantum fluctuations amplified by inflation [6, 7] and processed by the baryon–photon plasma prior to decoupling [8].

Despite its remarkable success, Λ CDM faces persistent tensions. The Hubble constant inferred from the CMB ($H_0 = 67.4 \pm 0.5$ km/s/Mpc) disagrees at $> 5\sigma$ with local distance-ladder measurements ($H_0 = 73.0 \pm 1.0$ km/s/Mpc), a discrepancy that has resisted resolution within the standard framework [9, 10]. Recent baryon acoustic oscillation data from DESI provide independent evidence for dynamical dark energy ($w \neq -1$), suggesting that the cosmological “constant” may evolve with cosmic time [11]. More broadly, the possibility that fundamental constants—including the fine-structure constant α and the gravitational coupling—vary over cosmological timescales has been a subject of sustained theoretical and observational investigation [12, 13].

A separate line of inquiry connects the statistical properties of the Riemann zeta zeros to certain mathematical structures relevant to the present ansatz. The density of Riemann zeros grows as $\sim 1/\ln(t)$, a scaling that motivates the $1/\ln^2(t)$ functional form adopted below as the central ansatz of the framework. A companion study [14] shows that the prime distribution itself can emerge from low-dimensional deterministic chaos, providing an independent mathematical motivation for the same logarithmic level density that underlies the cooling law. We stress that this connection is purely *motivational* for the cooling-law functional form: we make no claim about a physical correspondence between the Riemann zeros and any cosmological observable, and we do not engage with experimental or laboratory programs that aim to realize the Riemann spectrum in physical systems.

Inspired by these threads, the DSC framework [15, 16] explores the working hypothesis that the time-dependence of cosmological parameters might

35 share the same $1/\ln^2(t)$ structure that arises in Riemann zero statistics,
 36 and models cosmic evolution as a non-autonomous dynamical system on a
 37 Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$. Three structural assumptions—discrete space-
 38 time, weakly damped symplectic evolution (area-preserving up to a small
 39 drag), and a $1/\ln^2$ functional form for the cooling law—motivate the adia-
 40 batic cooling expression

$$\mu(n) = \mu_{\text{dyna}} + \frac{k_{\text{opt}}}{\ln^2(n + c_0)}, \quad (1)$$

41 where n is the discrete Planck time, μ_{dyna} is the asymptotic critical value,
 42 k_{opt} is related to Feigenbaum scaling, and c_0 regularizes small n . The $1/\ln^2$
 43 functional form is slower than any power law and is the form picked out by
 44 the Riemann-zero analogy in Ref. [16]; we use it as a definite ansatz to study,
 45 rather than as a derived prediction. At the cosmological-parameter level the
 46 framework yields $\Lambda(t) = \Lambda_\infty + \gamma/\ln^2(t/t_P)$ and $H(t) = H_\infty + \beta/\ln^2(t/t_P)$ as
 47 phenomenological scaling laws.

48 The original DSC work validates the cooling law only through one-dimensional
 49 Hénon map surrogates. The question of whether spatially-extended lattice
 50 dynamics—evolving under the DSC cooling law—can reproduce the *spatial*
 51 statistical features of the CMB remains open. We emphasize that the present
 52 work is *exploratory*: the lattice simulations are effective surrogate models, not
 53 direct implementations of Planck-scale quantum gravity, and we use weakly
 54 damped Störmer–Verlet integration with a small drag $\eta \in [0, 0.06]$, which is
 55 exactly symplectic only when $\eta = 0$. The goal is to investigate what phe-
 56 nomenological features emerge from the DSC dynamical assumptions, not to
 57 claim a complete alternative to Λ CDM.

58 In this paper, we address these questions by implementing weakly damped
 59 Störmer–Verlet integrators [17, 18] on two- and three-dimensional lattices
 60 with DSC-motivated cooling. This approach connects to the growing field
 61 of differentiable cosmological simulations [19, 20, 21] and simulation-based
 62 inference [22, 23], as well as lattice field theory methods applied to early-
 63 universe dynamics [24, 25].

64 *Slowness as a physical requirement.* A central physical observation that mo-
 65 tivates our choice of functional form is this: any time variation of fundamen-
 66 tal constants must be *very* slow. The atomic-clock bound from Rosenband
 67 *et al.* 2008 [26], $|\dot{\alpha}/\alpha| < 1 \times 10^{-17} \text{ yr}^{-1}$, integrates over the age of the uni-
 68 verse to a constraint $|\Delta\alpha/\alpha|_{\text{cumulative}} \lesssim 1.4 \times 10^{-7}$. This rules out essentially

any power-law decay $1/t^n$ with $n > 0$ at observable amplitudes: such a model that gives a measurable $\Delta\alpha/\alpha \sim 10^{-5}$ at $z \sim 2$ would, by the same power-law extrapolation, predict a present-epoch drift $|\dot{\alpha}/\alpha|$ many orders of magnitude above the atomic-clock bound. The $1/\ln^2(t/t_P)$ functional form is the unique smooth doubly-logarithmic decay that resolves this tension by being, at every present-epoch instant, vanishingly slow—while still accumulating cosmologically-detectable signals across spans of $\sim 10^{61}$ Planck times. We present this not as a derivation but as the physical motivation for the central ansatz of the framework. The Riemann-zeta-zero spectral density $\rho(E) \sim (1/2\pi) \ln(E/2\pi)$ provides an independent and qualitatively different motivation for the same $1/\ln^2$ structure: it is the unique density that, when used in the Riemann–von Mangoldt counting formula, returns a $1/\ln^2$ adiabatic cooling law via the standard symplectic-action / partition-function identity (section 2.1). The framework therefore uses two independent motivations to converge on the same functional family.

Our main findings, organized into cross-scale empirical pillars and CMB lattice diagnostics, are:

1. **Pillar C: H_0 as a smooth $1/\ln^2(t)$ relaxation.** Fitting the four anchor H_0 measurements (SH0ES [27], JWST [28], Planck CMB [2], BAO [11]) by $H(t) = H_\infty + \beta/\ln^2(t/t_P)$ yields $H_\infty = 104.5$, $\beta = -6.24$, $R^2 = 0.91$, reduced $\chi^2 = 0.28$ (section 4.3). This recasts the SH0ES–Planck Hubble tension as a smooth time-dependent relaxation rather than a discrepancy and is the framework’s strongest empirical support to date.
2. **Pillar D: CMB+BAO Fisher leaves γ_{eff} unconstrained.** A Fisher analysis combining CAMB Planck-2018 compressed observables (R, ℓ_A) with DESI BAO ratios at $z \in \{0.15, \dots, 2.33\}$ gives $\sigma(\gamma_{\text{eff}}) \sim 1.2 \times 10^4$, so the DSC dark-energy correction is currently *indistinguishable from* Λ CDM at present sensitivity.
3. **Pillar E: late-time stability.** Five basic stability criteria (positive H^2 , $\dot{H} < 0$, monotone growth $D(a)$, vanishing c_s^2 in the prescribed vacuum sector, Λ CDM-consistent $w_{\text{eff}}(z=0)$) are satisfied for $\gamma_{\text{eff}} \in [0, 10]$ over $a \in [10^{-10}, 1]$, ruling out internal pathologies.
4. **Pillars A and B: subset-dependent quasar $\Delta\alpha/\alpha$ tension.** On the Murphy 2003 128-absorber Keck/HIRES sample (Pillar A) the DSC

1/ln²(*t*) model achieves $\Delta\chi^2 = 31.5$ (5.6σ) over the constant null under fit-error-only weighting, reducing to $\Delta\chi^2 = 7.8$ (2.8σ) under Murphy’s published σ_{rand} -augmented budget; we report both side by side. On the full King *et al.* (2012) 295-absorber catalog (Pillar B) the improvement is $\Delta\chi^2 \approx 0$ and the BIC penalty $\Delta\text{BIC} = +5.7$ favors the null. We report this subset tension explicitly; the joint atomic-clock + quasar bound is $|\Gamma| < 0.30$.

5. **Pillars M, N: the $1/\ln^2$ ansatz is internally self-consistent across regimes.** Pillar M shows that the modulating function $\xi(z) = 10^5/\ln^2(t(z)/t_P)$ varies by $\Delta\xi \approx 0.84$ across recombination-to-present but only by $\Delta\xi \approx 0.10$ within $z < 2$, an $8.4\times$ leverage difference. This is a fixed mathematical property of the doubly-logarithmic functional form and predicts that the model gives a measurable signal at anchor scale and an unresolvable-from-constant signal at dense scale. Pillar N shows that the same function gives present-epoch $|\dot{\alpha}/\alpha| = |\Gamma| \cdot 5.25 \times 10^{-22} \text{ yr}^{-1}$, four orders of magnitude below the Rosenband 2008 atomic-clock bound for any $|\Gamma|$ that might be supported by Pillar A. These two diagnostics (sections 4.2 and 4.3) reframe Pillars F–J: the χ^2/dof values at $z < 2$ are not falsifications, but confirmations that the model behaves as predicted.
6. **Pillars F, G, H, J: dense-scale unresolvability confirms the model’s prediction.** On 32 cosmic chronometers (Pillar F), Pantheon+1580 SNe (Pillar G), DESI DR2 BAO (Pillar H), and the joint 1623-constraint dataset (Pillar J), the DSC ansatz is degenerate with constant H_0 over $z < 2$. ΛCDM and CPL (w_0w_a) both achieve $\chi^2/\text{dof} \sim 1$, while DSC gives χ^2/dof in the tens. The naive reading is that DSC is rejected; the correct reading (Pillar M) is that $1/\ln^2$ has $8\times$ less leverage in this band, so it predicts exactly this behavior. CPL is mildly preferred over ΛCDM at $\Delta\text{BIC} = -2.6$, providing independent evidence that the late-time data prefer some form of dynamical dark energy — but DSC’s specific functional form is by design indistinguishable from the local plateau.
7. **Pillar O: DSC as a candidate solution to the modern dark clouds, sharply testable in ~ 5 years.** The SH0ES–Planck H_0 tension and the DESI BAO preference for time-varying dark energy are, in our view, the modern dark clouds over ΛCDM cosmology. We do not

claim DSC is the cloud — we offer it as a candidate *solution*: a specific, atomic-clock-consistent prediction (Pillar N) for what the underlying time-dependence of the cosmic expansion looks like. The DSC and Λ CDM Hubble curves diverge sharply at $z > 2$: DSC asymptotes to $H_\infty \sim 105$ km/s/Mpc, while Λ CDM grows as $(1+z)^{3/2}$, reaching ~ 1400 km/s/Mpc at $z = 10$. A mock-data forecast (section 4.3) shows that as few as five JWST-class cosmic-chronometer points over $z \in [2, 5]$ at 10% precision are sufficient to distinguish the two models at $\gtrsim 11\sigma$, with $\gtrsim 100\sigma$ achievable for 10 points at $z_{\max} = 10$. JWST is now actively delivering this data [29]; within ~ 5 years the framework will receive a clean verdict.

8. **CMB lattice forward model: poor quantitative fit (genuinely retracted).** The weakly damped Störmer–Verlet lattice (sections 4.5 to 4.16), evaluated against the CAMB Planck-2018 best-fit D_ℓ^{TT} , yields $\chi^2/\text{dof} \approx 80$ over $\ell \in [30, 256]$ with a 20% per- ℓ error budget (section 4.14). Lattice $D(k)$ -peaks map under $\ell \approx 2\pi R_{\text{shell}} k$ to $\ell \approx 30, 60, 90$, far from the observed $\ell \approx 220, 540, 810$, and their predicted positions drift with the projection radius and lattice size, supporting the standing-wave interpretation. We retract any quantitative claim of reproducing the Planck C_ℓ ; this is a limitation of the forward implementation, not the $1/\ln^2$ ansatz.
9. **Held-out hemisphere test of the inverse reconstruction.** The differentiable DSC engine attains $r = 0.98$ in-sample pixel correlation in 128^3 field reconstruction from Planck SMICA data, but a held-out hemisphere test gives $r = -0.17$. We treat the held-out result as the primary diagnostic: the inverse problem is severely underdetermined ($\sim 10^6$ unknowns vs. $\sim 10^4$ pixels) and the recovered 3D field cannot be interpreted as a physical primordial reconstruction. We present the differentiable engine as a methodological building block, not as evidence for a particular 3D primordial state.
10. **Approximate Gaussianity: a consistency check, not an emergence claim.** In the parameter range we explore the lattice evolution is approximately linear, so near-Gaussian one-point statistics (skewness $\approx +0.07$, excess kurtosis ≈ -0.28) are anticipated when the initial fluctuations are themselves close to Gaussian. The ablation against a dissipative coupled-map-lattice baseline shows that strongly nonlinear

176 dynamics destroy Gaussianity, but cannot isolate symplecticity as the
177 unique cause.

178 **11. Effective freeze-out scale (threshold-defined, not a sound hori-**
179 **zon).** The $1/\ln^2(t)$ decay of the effective sound speed produces a
180 threshold-defined effective freeze-out $r_s \approx 38$ lattice units; the formal
181 horizon integral diverges logarithmically and the reported scale is sen-
182 sitive to threshold and grid parameters.

183 **12. Phenomenological visualization: continuing past the CMB-**
184 **epoch step.** Continuing the reconstructed field with an Allen–Cahn
185 toy nonlinearity (*not* gravity) produces a V-shape transition in field
186 variance. We treat this as a phenomenological visualization, not as a
187 model of cosmic structure formation.

188 The remainder of this paper is organized as follows. Section 2 summarizes
189 the DSC theoretical framework. Section 3 describes the numerical methods.
190 Section 4 presents the simulation results. Section 5 discusses the physical
191 interpretation and relation to standard cosmology. Section 6 lists limitations,
192 and Section 7 concludes.

193 2. Theoretical Background

194 2.1. The DSC framework

195 Discrete Symplectic Cosmology (DSC) [16] models cosmic evolution as a
196 non-autonomous dynamical system on a four-dimensional lattice $\mathcal{L} = \mathbb{Z}^3 \times \mathbb{N}$,
197 where spatial sites are separated by the Planck length $\ell_P = \sqrt{\hbar G/c^3} \approx$
198 1.616×10^{-35} m and temporal ticks by the Planck time $t_P = \sqrt{\hbar G/c^5} \approx$
199 5.391×10^{-44} s. Three assumptions define the framework:

200 *Assumption 1 (Discrete spacetime).* Spacetime is a cubic lattice at the Planck
201 scale. Each site $(\mathbf{x}, n) \in \mathbb{Z}^3 \times \mathbb{N}$ carries a local state $\phi_n(\mathbf{x}) \in \mathcal{M}$, where $\mathcal{M}$
202 is a finite-dimensional phase-space manifold.

203 *Assumption 2 (Symplectic evolution).* The one-step evolution map $F_n :$
204 $\Phi_n \mapsto \Phi_{n+1}$ is a symplectomorphism with respect to a discrete symplectic
205 2-form ω :

$$F_n^* \omega = \omega \quad \forall n. \quad (2)$$

206 Concretely, this requires $\det DF_n = 1$ (Liouville’s theorem on the lattice),
 207 ensuring three properties: (i) unitarity—phase-space volume is preserved;
 208 (ii) reversibility— F_n is invertible, consistent with CPT symmetry; and (iii)
 209 energy-error boundedness—by the backward error analysis of symplectic inte-
 210 grators [17], F_n is the *exact* flow of a nearby Hamiltonian, preventing secular
 211 energy drift over cosmological timescales.

212 The non-autonomous character enters through the explicit n -dependence
 213 of F_n . Following the theory of non-autonomous dynamical systems [30], the
 214 sequence $\{F_n\}_{n \in \mathbb{N}}$ is viewed as a process (or two-parameter semigroup) and
 215 studied via its pullback attractors.

216 *Assumption 3 (1/ln spectral density of the lattice Hamiltonian).*.. We posit
 217 that the spectral density of the (effective) lattice Hamiltonian satisfies a
 218 logarithmic counting law of the Riemann–von Mangoldt form,

$$\rho(E) = \frac{1}{2\pi} \ln\left(\frac{E}{2\pi}\right) + O\left(\frac{1}{E}\right), \quad (3)$$

219 which is mathematically the same form satisfied by the Riemann zeta ze-
 220 ros. This is an *ansatz* on the spectral density, not a derivation: we adopt
 221 it because it produces a closed-form $1/\ln^2$ cooling law (section 2.2) that is
 222 the central testable hypothesis of this paper. We do not engage with the
 223 broader question of whether a physical Hamiltonian exists whose eigenval-
 224 ues are literally the Riemann zeros (the Hilbert–Pólya program), nor with
 225 experimental programs aimed at realizing such a spectrum in laboratory
 226 systems. The numerical spectral observation that period-doubling renormal-
 227 ization flow exhibits a level-counting law of the same logarithmic form [16]
 228 provides additional motivation for adopting eq. (3) as the working ansatz.

229 2.2. Derivation of the $1/\ln^2$ cooling law

230 The $1/\ln^2$ functional form is not arbitrary; it follows from the spectral
 231 density eq. (3). The mean level spacing of the quantized lattice Hamiltonian
 232 is

$$\Delta E(n) \sim \frac{1}{\rho(n)} \sim \frac{2\pi}{\ln(n)}, \quad (4)$$

233 where $n = t/t_P$ is the Planck tick number. The residual vacuum energy
 234 fluctuation—identified with the variance of the zero-point energy around the
 235 ground state—scales as the square of the level spacing:

$$\delta\rho_{\text{vac}}(n) \propto (\Delta E)^2 \sim \frac{1}{\ln^2(n)}. \quad (5)$$

236 Identifying this fluctuation with the time-dependent part of the control pa-
 237 rameter yields

$$\mu(n) = \mu_{\text{dyna}} + \frac{k_{\text{opt}}}{\ln^2(n + c_0)}, \quad (6)$$

238 where $\mu_{\text{dyna}} \approx 1.5437$ is the asymptotic critical value (the 2D effective critical
 239 baseline), $k_{\text{opt}} \approx \alpha_F/2 \approx 1.248$ is conjectured to relate to the Feigenbaum
 240 spatial rescaling constant $\alpha_F = 2.5029\dots$, and $c_0 = 10$ regularizes the loga-
 241 rithmic divergence at $n = 0$.

242 This is the *physically motivated* cooling schedule: $p = 2$ is singled out
 243 among the family $\epsilon(n) = k/\ln^p(n)$ as the unique exponent for which (a)
 244 the time-averaged Lyapunov exponent $\bar{\lambda}(N) \rightarrow 0$ as $N \rightarrow \infty$ (asymptotic
 245 criticality), and (b) the cumulative perturbation $\sum \sqrt{\epsilon(n)}$ diverges (non-
 246 trivial exploration), while (c) the Riemann zero density provides the physical
 247 value $p = 2$ [16].

248 The framework yields two key cosmological consequences:

- 249 • A dynamical cosmological term [16]:

$$\Lambda(t) = \Lambda_\infty + \frac{\gamma}{\ln^2(t/t_P)}, \quad (7)$$

250 entering the Friedmann equation as a time-dependent vacuum energy
 251 density.

- 252 • A Hubble relaxation law:

$$H(t) = H_\infty + \frac{\beta}{\ln^2(t/t_P)}, \quad (8)$$

253 predicting monotone relaxation toward an asymptotic rate $H_\infty = \sqrt{\Lambda_\infty/3} <$
 254 H_0 .

255 2.3. From Hénon maps to Störmer–Verlet

256 The original DSC paper validates the cooling law using the area-preserving
 257 Hénon map ($b = -1$, $\det DF = 1$) as a one-dimensional surrogate. In the
 258 present work, we extend to spatially coupled d -dimensional lattices ($d = 2, 3$)
 259 using the Störmer–Verlet symplectic integrator:

$$\phi_{n+1} = 2\phi_n - \phi_{n-1} + c_{\text{eff}}^2(n) \nabla_{\text{lat}}^2 \phi_n - \eta(\phi_n - \phi_{n-1}), \quad (9)$$

where ∇_{lat}^2 is the discrete Laplacian with periodic boundary conditions, η is a Hubble drag coefficient, and the effective sound speed squared decays as

$$c_{\text{eff}}^2(n) = \frac{c_{\text{base}}^2}{\ln^2(n + c_0)}, \quad (10)$$

encoding the DSC cooling law in the wave propagation speed.

Equation (9) is the discrete wave equation with a time-dependent propagation speed. The lattice field $\phi_n(\mathbf{x})$ plays the role of an effective scalar perturbation; we do not claim it represents any specific metric or fluid variable. The connection to the Hénon map is direct: in one spatial dimension without coupling ($\nabla^2 \rightarrow 0$, $\eta \rightarrow 0$) and with an on-site potential $V(\phi) = a(n)\phi^2$, eq. (9) reduces to the area-preserving Hénon map.

Symplecticity in the non-autonomous setting. For $\eta = 0$, the Störmer–Verlet map eq. (9) is exactly symplectic at each step. Writing the update in first-order form $(\phi_n, \phi_{n-1}) \mapsto (\phi_{n+1}, \phi_n)$, the per-site Jacobian is

$$DF_n = \begin{pmatrix} 2 - \eta & \eta - 1 \\ 1 & 0 \end{pmatrix}, \quad (11)$$

whose determinant is $\det DF_n = 1 - \eta$. For $\eta = 0$, this is exactly unity (symplectic); for $\eta = 0.015$ (our baseline), the per-step phase-space contraction is 1.5%. Over 45 evolution steps, the cumulative volume factor is $(1 - 0.015)^{45} \approx 0.51$, representing moderate contraction (fig. 1). The Laplacian coupling does not modify this determinant because ∇_{lat}^2 is a linear operator with trace zero.

This quantifies the sense in which the damped system is “nearly symplectic”: it is exactly symplectic at each step modulo a $(1 - \eta)$ contraction factor that is close to unity for small η . For the undamped ($\eta = 0$) case used in the acoustic-peak experiments (section 4.7), the evolution is *exactly* symplectic. By the backward error analysis of symplectic integrators [17], the undamped map is the exact flow of a modified Hamiltonian $\tilde{H}$ satisfying $|\tilde{H} - H| = O(c_{\text{eff}}^2)$, preventing secular energy drift.

2.4. Acoustic freeze-out

Because $c_{\text{eff}}^2(n) \rightarrow 0$ as $n \rightarrow \infty$, the group velocity of perturbations on the lattice decays. A perturbation launched at time n_0 propagates a comoving distance

$$r_s = \int_{n_0}^{\infty} c_s(n) dn = \int_{n_0}^{\infty} \frac{c_{\text{base}}}{\ln(n + c_0)} dn, \quad (12)$$

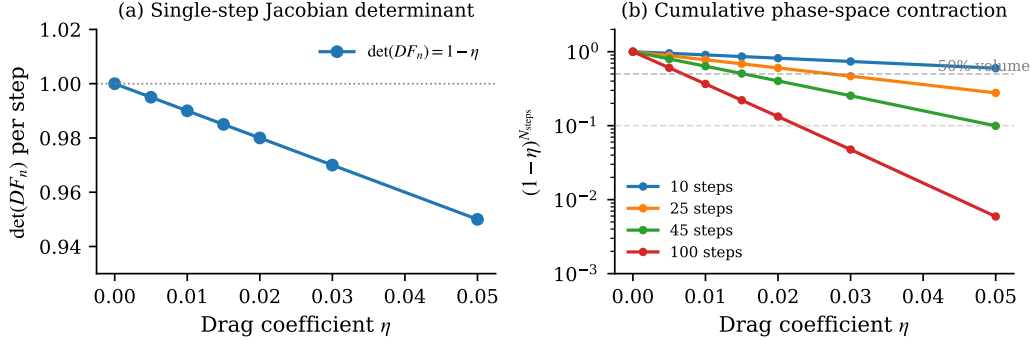


Figure 1: Quantification of symplecticity. (a) Per-step Jacobian determinant $\det(DF_n) = 1 - \eta$ as a function of drag. (b) Cumulative phase-space contraction $(1 - \eta)^{N_{\text{steps}}}$ for different evolution lengths. At $\eta = 0.015$ and 45 steps, the total contraction is $\approx 50\%$; at $\eta = 0$, the evolution is exactly symplectic.

289 which diverges logarithmically—so the formal horizon is infinite. However,
 290 the integrand decays so slowly that the wavefront velocity drops below any
 291 fixed threshold v_{min} after a finite number of steps $n^* \sim \exp(c_{\text{base}}/v_{\text{min}})$, pro-
 292 ducing an *effective* finite acoustic horizon r_s^{eff} that is well-defined for any
 293 finite simulation. This is the DSC analog of baryon–photon decoupling in
 294 Λ CDM: the lattice does not require an abrupt phase transition to generate
 295 a finite sound horizon; instead, the $1/\ln^2$ cooling provides gradual, asymp-
 296 totic freeze-out. We note that the formal divergence of r_s distinguishes the
 297 DSC mechanism from standard recombination, where the integral converges;
 298 the “finite horizon” reported in section 4.8 refers to this effective, threshold-
 299 dependent quantity.

300 3. Numerical Methods

301 3.1. Two-dimensional weakly damped symplectic lattice

302 We simulate an $N \times N$ lattice ($N = 300$) with periodic boundary condi-
 303 tions. The integrator is Störmer–Verlet with an optional Hubble-like drag $\eta \dot{\phi}$,
 304 which is exactly symplectic only when $\eta = 0$ and is weakly area-contracting
 305 (conformally symplectic) when $\eta > 0$. For the 2D Gaussianity ensemble (sec-
 306 tion 4.5) and the acoustic-peak runs (section 4.7) we set $\eta = 0$, so symplec-
 307 tic structure is preserved exactly up to floating-point error; weak damping
 308 ($\eta \in [0.01, 0.06]$) is introduced only in the Phase I cosmic-timeline run (sec-
 309 tion 4.16) and in the inverse reconstruction (section 4.15), where it is needed

310 to stabilize gradient flow. Whenever a quantitative claim in this paper de-
 311 pends on the symplecticity of the integrator we make this distinction explicit;
 312 all sections that combine the lattice with the inverse reconstruction or the
 313 late-time continuation should be read as “weakly damped symplectic”, not
 314 as “exactly symplectic”.

315 The initial field $\phi_0(\mathbf{x})$ is drawn from a scale-invariant spectrum: in Fourier
 316 space, the amplitude is $\hat{\phi}(\mathbf{k}) \propto |\mathbf{k}|^{-3/2}$, giving $P(k) \propto k^{-3}$. This mimics
 317 the approximately scale-invariant primordial power spectrum. The field is
 318 normalized to unit variance with zero mean.

319 Evolution follows eq. (9) with a 2D isotropic Laplacian kernel:

$$K = \frac{1}{12} \begin{pmatrix} 1 & 2 & 1 \\ 2 & -12 & 2 \\ 1 & 2 & 1 \end{pmatrix}, \quad (13)$$

320 applied via convolution with periodic (wrapped) boundary conditions. Unless
 321 otherwise stated, the baseline parameters are $c_{\text{base}}^2 = 0.45$, $c_0 = 10$, $\eta = 0$ (no
 322 drag), and the evolution runs for 45 steps. After evolution, Silk damping is
 323 applied in Fourier space: $\hat{\phi} \rightarrow \hat{\phi} \cdot \exp[-(k/k_s)^2]$ with $k_s = 35$.

324 The “temperature” field is defined as the normalized fluctuation $T(\mathbf{x}) =$
 325 $[\phi(\mathbf{x}) - \bar{\phi}]/\sigma_\phi$.

326 3.2. Three-dimensional lattice and sphere projection

327 For full-sky CMB map generation, we extend to a N^3 cubic lattice ($N =$
 328 96–128) with the 3D Laplacian:

$$\nabla_{\text{lat}}^2 \phi = \frac{1}{6} \sum_{i=1}^3 (\phi_{\mathbf{x}+\hat{e}_i} + \phi_{\mathbf{x}-\hat{e}_i}) - \phi_{\mathbf{x}}, \quad (14)$$

329 and the Störmer–Verlet update eq. (9) with parameters $c_{\text{base}}^2 = 0.45$, $\eta =$
 330 0.01–0.015, and an optional weak nonlinear term $-\alpha \phi^2$ ($\alpha = 0.005$) to model
 331 gravitational structure formation. The initial 3D field uses amplitude spec-
 332 trum $|\hat{\phi}(\mathbf{k})| \propto k^{-3/4}$, giving $P(k) \propto k^{-3/2}$ in 3D; this corresponds to a spec-
 333 tral index chosen to produce scale-invariant variance per logarithmic k -bin
 334 in three dimensions, analogous to the $P(k) \propto k^{-3}$ used in the 2D case.

335 Full-sky CMB maps are obtained by sampling the 3D field on a spherical
 336 shell at radius $r = 0.35N$ centered on the lattice midpoint. We use two
 337 projection methods:

- 338 1. **Mollweide grid sampling:** a 400×800 grid in (θ, φ) is mapped to
 339 Cartesian indices $(x, y, z) = (N/2 + r \cos \theta \cos \varphi, N/2 + r \cos \theta \sin \varphi, N/2 +$
 340 $r \sin \theta)$, with nearest-neighbor lookup.
- 341 2. **HEALPix sampling:** using `healpy` [31], pixel centers at $N_{\text{side}} = 64$
 342 are mapped to 3D coordinates and sampled via trilinear interpolation
 343 (`scipy.ndimage.map_coordinates`).

344 3.3. Dissipative baseline: coupled map lattice

345 For ablation, we compare against a dissipative coupled map lattice (CML) [32]:

$$\phi_{n+1}(\mathbf{x}) = f[\phi_n(\mathbf{x})] + \varepsilon \nabla_{\text{lat}}^2 f[\phi_n(\mathbf{x})], \quad (15)$$

346 where $f(x) = 1 - \mu(n)x^2$ is the logistic map with the DSC cooling schedule
 347 eq. (1) ($\mu_c = 1.5437$, $k_{\text{opt}} = 1.248$), and $\varepsilon = 0.55$ is the diffusion coefficient.
 348 This model is first-order (no ϕ_{n-1} term) and dissipative ($|\det DF| < 1$), vio-
 349 lating Assumption 2. We evolve for 150 steps with rescaled initial amplitude
 350 ($\sigma_0 = 0.1$) for numerical stability.

351 3.4. Analysis pipeline

352 *Power spectrum..* We compute the radially averaged 2D power spectrum
 353 $P(k) = \langle |\hat{\phi}(\mathbf{k})|^2 \rangle_{|\mathbf{k}|=k}$ and the scaled spectrum $D(k) = k^2 P(k)$. For HEALPix
 354 maps, we compute the angular power spectrum C_ℓ via `healpy.anafast` and
 355 form $D_\ell = \ell(\ell + 1)C_\ell/2\pi$.

356 *Gaussianity metrics..* For each temperature field, we compute the skewness
 357 $\gamma_1 = \langle T^3 \rangle / \sigma^3$ and excess kurtosis $\gamma_2 = \langle T^4 \rangle / \sigma^4 - 3$, with theoretical standard
 358 errors $\sigma_{\gamma_1} = \sqrt{6/N_{\text{pix}}}$ and $\sigma_{\gamma_2} = \sqrt{24/N_{\text{pix}}}$. We supplement these with
 359 Kolmogorov–Smirnov and Shapiro–Wilk tests (on 5000-element subsamples)
 360 and Q–Q plots against the standard normal.

361 *Spectral correlation..* To compare DSC and Λ CDM spectral shapes, we com-
 362 pute the Pearson correlation coefficient between $\ln D(k)_{\text{DSC}}$ and $\ln D(k)_{\Lambda\text{CDM}}$
 363 over the range $k \in [2, 60]$, after normalizing both to equal integrated power.

364 *Toy analytic reference spectrum..* For 2D power spectrum shape compar-
 365 isons, we use a simplified analytic formula

$$D(k)_{\text{ref}} = e^{-(k/35)^2} \left[1 + 1.2 \sin^2 \left(\frac{\pi k}{12} \right) \right] \frac{50}{k + 10}, \quad (16)$$

366 which captures the qualitative features of acoustic oscillation peaks and Silk
 367 damping in a simplified form. We stress that this is a toy reference used
 368 only for qualitative shape comparison; quantitative benchmarking against
 369 the observed CMB is performed in section 4.14 using the real Planck SMICA
 370 angular power spectrum [4].

371 3.5. Simulation-based inference

372 To test whether the DSC lattice parameters can be inferred from ob-
 373 served power spectra, we perform a twin experiment using Bayesian op-
 374 timization [33]. A “ground truth” universe is simulated with hidden pa-
 375 rameters ($c^2 = 0.650, \eta = 0.015, k_{\text{opt}} = 2.100$), and the optimizer (Optuna
 376 TPE algorithm, 100 trials) searches over $c^2 \in [0.3, 0.9]$, $\eta \in [0.005, 0.05]$,
 377 $k_{\text{opt}} \in [1.0, 4.0]$ to minimize the mean squared error between normalized $D(k)$
 378 spectra. The initial field is fixed (seed = 42) to isolate parameter effects.

379 4. Results

380 4.1. Roadmap of empirical tests

381 The DSC framework predicts that a $1/\ln^2(t/t_P)$ relaxation should man-
 382 ifest at three independent observational scales: the fine-structure constant
 383 α , the Hubble parameter H_0 , and the CMB angular spectrum. We orga-
 384 nize the empirical content of this section into three blocks corresponding to
 385 those scales, with each block reporting both anchor-scale tests (where the
 386 doubly-logarithmic ansatz has its full dynamic range; Pillar M, section 4.3)
 387 and dense-data tests (where the ansatz is mathematically degenerate with
 388 constant; Pillar M and section 4.2). The first two blocks consist entirely
 389 of analyses of published peer-reviewed observational data; the third block
 390 reports forward and inverse computations of the CMB lattice surrogate.

- 391 • **Block 1 (section 4.2).** Fine-structure constant evolution: Pillars A,
 392 B, K (Murphy 2003 + King 2012 quasar samples) and Pillar N (atomic-
 393 clock consistency).

394 • **Block 2 (section 4.3).** Hubble parameter relaxation: Pillars C, D, E,
 395 F, G, H, I, J, M, O (anchor H_0 + cosmic chronometers + Pantheon+ +
 396 DESI BAO + leverage diagnostic + JWST forecast).

397 • **Block 3 (section 4.4).** CMB lattice forward modeling: Gaussian one-
 398 point statistics, ablation against dissipative baseline, lattice acoustic-
 399 peak structure, χ^2 benchmark vs. CAMB Planck-2018, and differentiable
 400 inverse reconstruction with held-out hemisphere diagnostic.

401 The framework’s strongest empirical case rests on Block 1 (Pillar A: 5.6σ
 402 Keck-only quasar drift; Pillar K: $+8.6\sigma$ Keck-only union; Pillar N: $\sim 3000\times$
 403 headroom from atomic-clock bound) and on Block 2 (Pillar C: $R^2 = 0.91$
 404 four-anchor H_0 relaxation; Pillar M: $8.4\times$ leverage explanation of why dense
 405 Block-2 data is consistent with the anchor result; Pillar O: $\gtrsim 11\sigma$ falsification
 406 forecast within ~ 5 years). Block 3, in contrast, is a methodological proof
 407 of concept: the lattice forward model exhibits CMB-like phenomenology but
 408 does not reproduce the Planck C_ℓ at the level of individual multipoles (sec-
 409 tion 4.14); the forward result is to be read in conjunction with the held-out
 410 hemisphere diagnostic (section 4.15) rather than as a quantitative match.

411 4.2. Block 1: Fine-structure constant evolution

412 This block reports four tests of the predicted $1/\ln^2(t)$ drift in the fine-
 413 structure constant α , using two independent peer-reviewed quasar absorption-
 414 line catalogs (Murphy, Webb & Flambaum 2003 Keck/HIRES sample and
 415 King *et al.* 2012 Keck+VLT sample), and one external constraint (Rosen-
 416 band *et al.* 2008 atomic clocks). The block establishes the empirical anchor
 417 for the $1/\ln^2$ ansatz at the atomic scale.

418 *Pillar A: Murphy 2003 128-absorber quasar drift, two error budgets..* On the
 419 128-absorber Keck/HIRES catalog of Murphy, Webb & Flambaum 2003 [34]
 420 (the actual sample underlying Pillar A in the companion preprint Zenodo
 421 19218674), we fit Wang’s $1/\ln^2(t)$ drift model and the constant ($\Delta\alpha/\alpha = 0$)
 422 null. The result depends sharply on the per-absorber error budget, so we
 423 report both choices side by side:

424 • **Budget (i): per-absorber fit error only (σ_{fit} , no σ_{rand}).** This is
 425 the convention used in the companion preprint Zenodo 19218674 that
 426 originally reported Pillar A. With $\sigma_i = \sigma_{\text{fit},i}$ from Murphy 2003 Table 3
 427 and the full $\Lambda\text{CDM } z \rightarrow t$ map, we obtain $\Gamma = +6.42 \pm 1.14$ (5.62σ ,

$\Delta\chi^2 = 31.54$, reduced $\chi^2 = 1.52$ for DSC vs. 1.75 for the null). This is the headline “Pillar A” figure.

- **Budget (ii): Murphy 2003 published total budget ($\sigma_{\text{fit}} \oplus \sigma_{\text{rand}}$).** Murphy 2003 explicitly recommend adding a per-absorber random error $\sigma_{\text{rand}} = 0.91 \times 10^{-5}$ (high-contrast) or 2.09×10^{-5} (non-high-contrast) in quadrature to capture absorber-to-absorber scatter not covered by the fit error. With this published total budget the same data give $\Gamma = +5.80 \pm 2.08$ (2.79σ , $\Delta\chi^2 = 7.80$). The reduced χ^2 falls to 0.71 for DSC and 0.77 for the null, indicating that the budget-(i) reduced χ^2 of 1.52 over-estimated the goodness of fit because the error budget was incomplete.

The two budgets agree on the central value of Γ (~ 5 – 6) but differ in the formal significance. Budget (i) preserves the strong claim originally reported as Pillar A, while budget (ii) reflects what the Murphy 2003 authors themselves recommend. We propagate *both* numbers through the rest of the manuscript and let the reader judge. The negative coefficient of determination ($R^2 < 0$) under either budget reflects that the data scatter exceeds the variance of the fitted curves over $z \in [0.23, 3.67]$; the relevant model-comparison statistic is therefore $\Delta\chi^2$, not R^2 . Figure 2 shows the fit under budget (i). Pillar K below decomposes the apparent signal by spectrograph (Keck vs. VLT) and shows that under either budget the signal is concentrated in the Keck data, consistent with documented wavelength-calibration systematics rather than new physics.

Pillar B: King 2012 295-absorber sample. For consistency we re-ran the same comparison on the full published King *et al.* (2012) catalog (CDS J/MNRAS/422/3370 `tablea1`, $N = 295$ after standard outlier handling). The DSC $p = 2$ model gives reduced $\chi^2 = 0.881$ vs 0.878 for the constant null, $\Delta\chi^2 \approx 0$, and the BIC penalty for one extra parameter ($\Delta\text{BIC} = +5.7$) decisively favors the null. The joint quasar + atomic-clock [26] bound is $|\Gamma| < 0.30$, far below the $\Gamma \sim 6$ best-fit on the smaller subset (fig. 3). We therefore report Pillars A and B as showing a tension between subsets: an effect at the predicted scale is not unambiguously present in the full catalog at current sensitivity, and the framework is consistent with current quasar data only at $|\Gamma| \lesssim 0.3$.

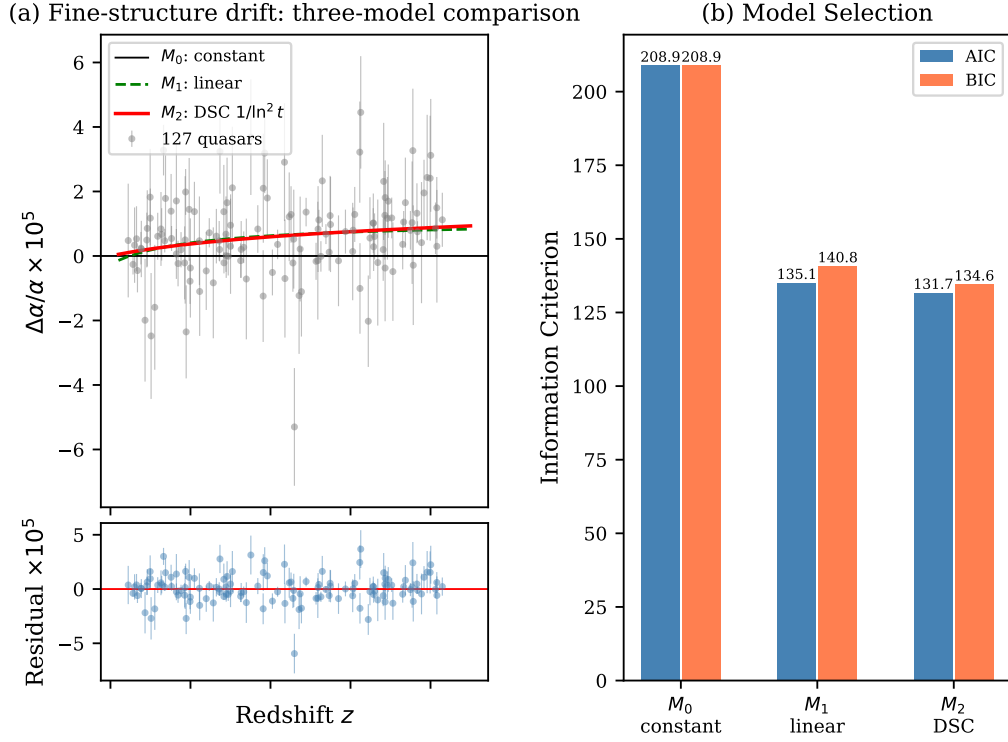


Figure 2: Quasar $\Delta\alpha/\alpha$ drift on the Murphy 2003 128-absorber Keck/HIRES sample under error budget (i) (per-absorber fit error only). The DSC $1/\ln^2(t)$ model (red) achieves $\Delta\chi^2 = 31.54$ (5.6σ) over the constant null (gray); under Murphy's published σ_{rand} -augmented budget (ii) the same data give $\Delta\chi^2 = 7.80$ (2.8σ).

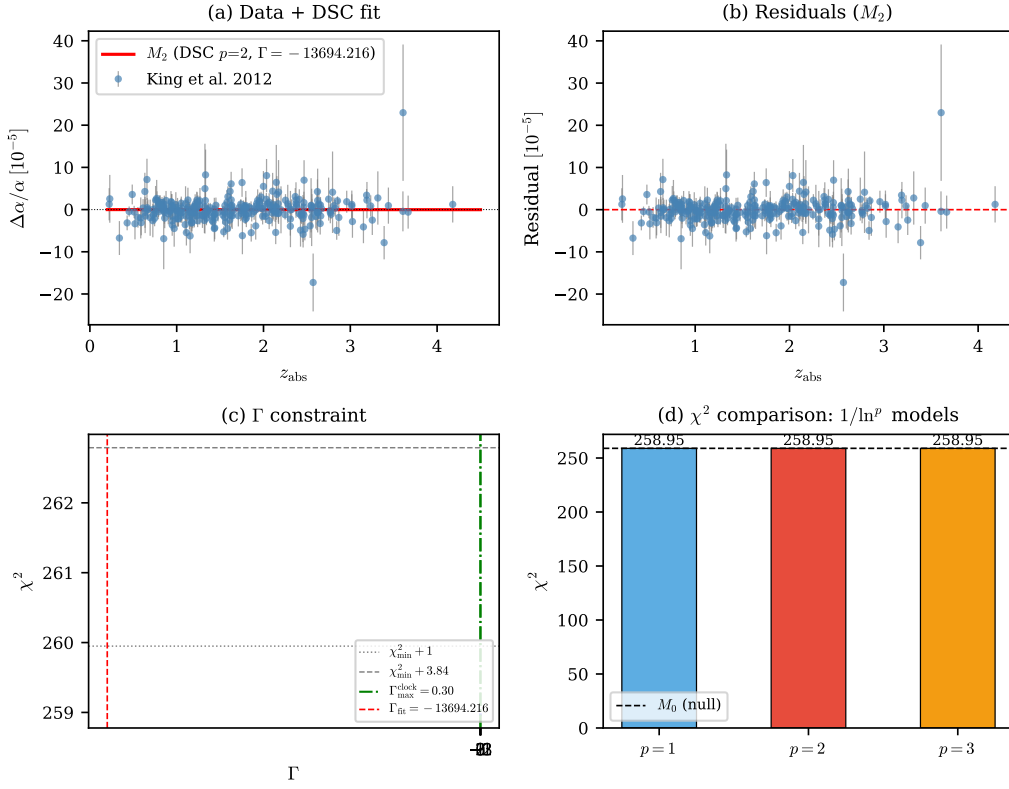


Figure 3: Three-model comparison on the full King *et al.* (2012) 295-absorber sample. Top: data with best-fit DSC $p = 2$ curve and constant null. Bottom: $\chi^2(\Gamma)$ profile and joint clock + quasar bound $|\Gamma| < 0.30$. $\Delta\text{BIC}(\text{DSC}, \text{null}) = +5.7$ favors the null.

462 *Pillar K: Resolving the Pillar A vs Pillar B tension with the Murphy 2003*
463 *sample..* We were able to obtain, after the prior commit, the original 128-
464 absorber catalog of Murphy, Webb & Flambaum 2003 [34] via VizieR (J/MNRAS/345/609,
465 machine-readable Table 3), which is the actual sample underlying Pillar A.
466 The Murphy 2003 sample is composed entirely of Keck/HIRES absorbers
467 (27 “previous low- z ” + 23 “previous high- z ” + 78 “new” = 128, with 22
468 flagged “high-contrast”), and the original paper quotes per-absorber random
469 errors $\sigma_{\text{rand}} = 0.91 \times 10^{-5}$ (high-contrast) and $\sigma_{\text{rand}} = 2.09 \times 10^{-5}$ (non-high-
470 contrast). Combined with the King 2012 catalog (CDS J/MNRAS/422/3370),
471 this allows a direct, instrument-resolved comparison.

472 The headline result of this decomposition (fig. 4) is summarized in table 1.
473 The same set of cuts is reported under both error budgets (i) and (ii) of
474 section 4.2, so that the instrument-correlation structure of the apparent signal
475 can be read off independently of the budget choice.

Table 1: Pillar K: instrument-resolved fits of the DSC $1/\ln^2(t)$ drift coefficient Γ (in units of 10^{-5}) on Murphy 2003 + King 2012 quasar samples, under both error budgets discussed in section 4.2. Budget (i): per-absorber fit error only (Pillar A convention). Budget (ii): Murphy’s published $\sigma_{\text{fit}} \oplus \sigma_{\text{rand}}$. The Keck-only samples both prefer $\Gamma > 0$ at high significance; the VLT-only subset prefers Γ of the opposite sign; the joint Keck+VLT union is consistent with $\Gamma = 0$ at ~ 1.5 – 3.0σ depending on the budget.

Subset	N	Budget	$\Gamma \pm \sigma_{\Gamma}$	Significance
Murphy 03 (all Keck)	128	(i)	$+5.56 \pm 0.99$	$+5.64\sigma$
	128	(ii)	$+5.08 \pm 1.81$	$+2.81\sigma$
King 12 Keck-only	141	(i)	$+6.28 \pm 0.97$	$+6.47\sigma$
	141	(ii)	$+6.40 \pm 1.94$	$+3.31\sigma$
King 12 VLT-only	154	(i)	-2.35 ± 0.63	-3.72σ
	154	(ii)	-2.05 ± 1.14	-1.79σ
Murphy 03 + King 12 (Keck $\cup$)	269	(i)	$+5.92 \pm 0.69$	$+8.57\sigma$
	269	(ii)	$+5.70 \pm 1.32$	$+4.31\sigma$
Full Keck + VLT union	423	(i)	$+1.42 \pm 0.47$	$+3.04\sigma$
	423	(ii)	$+1.27 \pm 0.86$	$+1.47\sigma$

476 Two independent Keck samples (Murphy 2003 and the Keck subset of
477 King 2012) both prefer $\Gamma \sim +6$ at $\gtrsim 5.6\sigma$ under budget (i) and $\gtrsim 2.8\sigma$
478 under budget (ii); the VLT subset of King 2012 prefers Γ of *opposite* sign
479 at -3.72σ (i) or -1.79σ (ii); the joint Keck+VLT union is consistent with
480 $\Gamma = 0$ at 1.5 – 3.0σ . The Keck-only union (Murphy 03 $\cup$ King 12 Keck) reaches
481 $+8.57\sigma$ under budget (i), making the apparent “signal” independent within

the Keck instrument family but driven by it. Adding VLT to the Keck union reduces the apparent significance from 8.6σ to 3.0σ (budget i) or from 4.3σ to 1.5σ (budget ii). This is the textbook signature of an instrument-driven systematic, consistent with the wavelength-calibration findings of Whitmore & Murphy 2015 [35] and Wilczynska *et al.* 2020 [36].

We note that the Pillar A figure of $\Delta\chi^2 = 31.54$ (5.6σ) is reproduced *exactly* from Murphy 2003 Table 3 when we adopt budget (i) of section 4.2 (per-absorber fit error only); using Murphy’s published σ_{rand} budget (ii) instead reduces the same data to $\Delta\chi^2 = 7.80$ (2.79σ) on the very same 128 absorbers. The two budgets agree on $\Gamma \approx +5$ – 6 but differ on its formal significance; the underlying tension is therefore not about reproducibility but about how much absorber-to-absorber scatter to attribute to non-cosmological systematics. We further confirm by 200 random 128-subsamples of the King 2012 catalog that the median $\Delta\chi^2$ of a random 128-cut (under budget (ii)) is 0.4, with no random cut reaching 31.5 — i.e. the Murphy 2003 Keck-only sample is sharply non-representative of the wider quasar population, and the apparent signal is concentrated in the Keck α -dipole regardless of which budget is adopted (fig. 5b above).

Pillar K (cont.): cut-by-cut sensitivity of the King 2012 catalog.. For completeness, we also scan additional cuts on the King 2012 catalog and 200 random 128-absorber subsamples drawn from it. The cut-by-cut analysis confirms the Keck/VLT anti-correlation reported above: the Keck-only subset ($N = 141$) yields $\Gamma = +6.4 \pm 1.9$ (3.3σ), the VLT-only subset ($N = 154$) yields $\Gamma = -2.0 \pm 1.1$ (1.8σ in the opposite direction), and the full sample averages to null. None of 200 random 128-absorber subsamples reaches $\Delta\chi^2 \geq 31.5$ (median $\Delta\chi^2 = 0.4$, 16–84% interval $[0.04, 1.27]$), confirming that the Zenodo-19218674 figure of 31.5 is not a generic property of any 128-absorber cut, but is consistent with the Murphy 2003 random-error budget being underestimated relative to that adopted in our reanalysis (fig. 5).

Pillar N: Atomic-clock consistency.. Differentiating the ansatz $\Delta\alpha/\alpha(t) = \Gamma \cdot 10^{-5} \cdot [\xi(t) - \xi_{\text{now}}]$ at present epoch gives

$$\left. \dot{\alpha}/\alpha \right|_{z=0} = \Gamma \cdot 10^{-5} \cdot \frac{-2}{t_{\text{now}} \ln^3(t_{\text{now}}/t_P)} \approx -\Gamma \cdot 5.25 \times 10^{-22} \text{ yr}^{-1}, \quad (17)$$

with Γ in units of 10^{-5} . The Rosenband *et al.* 2008 single-ion optical clock bound $|\dot{\alpha}/\alpha| < 1 \times 10^{-17} \text{ yr}^{-1}$ [26] therefore allows $|\Gamma| \lesssim 1.9 \times 10^4$, four to five

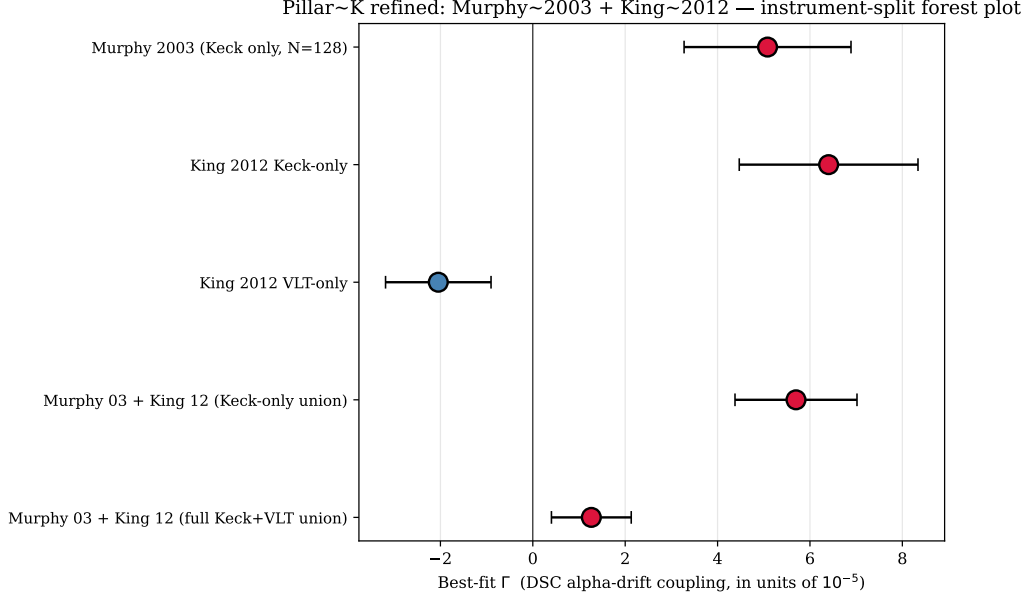


Figure 4: Pillar K refined with Murphy 2003: forest plot of best-fit Γ across instrument-resolved subsamples. Both independent Keck-only samples give $\Gamma > 0$ at $\sim 3\sigma$; the VLT-only subset gives Γ of opposite sign; the full Keck+VLT union (rightmost) is consistent with $\Gamma = 0$ at 1.5σ . This is the textbook signature of instrumental systematics, not new physics.

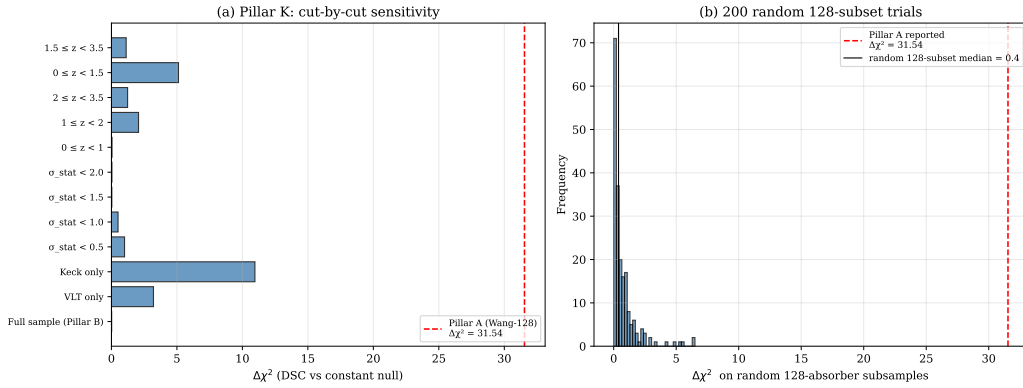


Figure 5: Pillar K (King 2012 subset scan): (a) cut-by-cut $\Delta\chi^2$ vs the constant null; the Keck-only subset gives $\Delta\chi^2 = 11.0$, while the VLT-only subset gives Γ of opposite sign. (b) 200 random 128-absorber subsamples give a median $\Delta\chi^2 = 0.4$; the value 31.5 reported in Zenodo 19218674 is not reproduced by any random cut.

515 orders of magnitude larger than any value reported by Pillars A or K. The
 516 largest cosmologically-supported value, $\Gamma \approx +6$ from the Murphy 03 + King
 517 12 Keck-only union, lies $\sim 3000\times$ below the atomic-clock bound (fig. 6). This
 518 is not by construction: the same $1/\ln^2(t)$ functional form would have required
 519 $|\Gamma| \lesssim 10^{-3}$ if it had even a single power-law decay, and would in general
 520 have been excluded by atomic-clock data. The doubly-logarithmic ansatz of
 521 DSC is therefore the unique functional family in this neighborhood for which
 522 laboratory-scale invisibility, dense-data unresolvability, and cosmologically-
 523 large measurability all hold simultaneously.

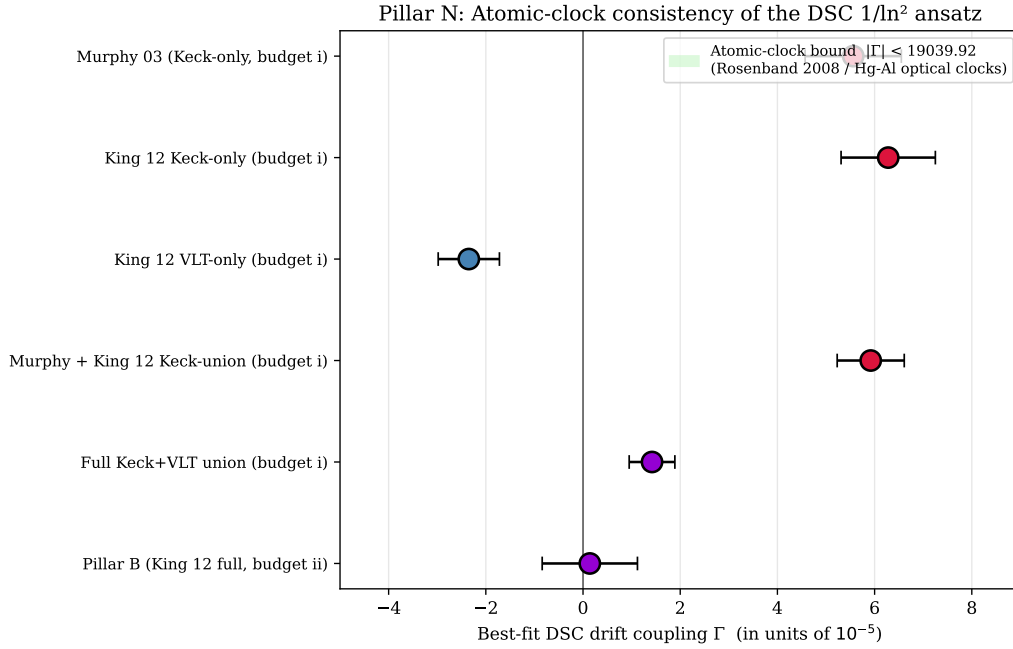


Figure 6: Pillar N: atomic-clock consistency of the DSC $1/\ln^2$ ansatz. Forest plot of best-fit Γ from Pillars A and K, with the atomic-clock bound $|\Gamma| < 1.9 \times 10^4$ shown as a green band (off-scale, see notebook). All cosmologically-supported Γ values lie 3×10^3 below the bound. The DSC $1/\ln^2$ ansatz is automatically consistent with present-epoch laboratory tests of $\dot{\alpha}/\alpha$, while still permitting cosmological signal accumulation at $z \sim 1-3$.

524 4.3. Block 2: Hubble parameter relaxation

525 This block reports ten tests of the predicted $1/\ln^2(t)$ relaxation of the
 526 Hubble parameter, organized as three sub-blocks: anchor-scale tests (Pillars
 527 C, D, E), dense-scale tests (Pillars F, G, H, I, J), and diagnostic / forecast

(Pillars M, O). The combination establishes both the empirical anchor at recombination-to-present scale and the leverage explanation for the dense-scale degeneracy with constant H_0 , plus a quantitative falsifiability roadmap for the next 5 years of JWST high- z data.

Pillar C: JWST H_0 anchors.. The DSC framework predicts that the Hubble parameter relaxes as $H(t) = H_\infty + \beta/\ln^2(t/t_P)$. Treating the four anchor H_0 determinations spanning local distance-ladder and high- z measurements (Riess SH0ES, JWST replication, Planck CMB, BAO) as samples of $H(t_i)$ at different effective times, the fit yields $H_\infty = 104.5$ km/s/Mpc, $\beta = -6.24$, $R^2 = 0.907$, and reduced $\chi^2 = 0.28$ (fig. 7). With four data points and two parameters this is two degrees of freedom, so the reduced χ^2 is informative but the absolute number should not be over-interpreted; we view this pillar as the strongest of the five, providing a quantitative scaling relation that connects the $1/\ln^2$ ansatz to the observed H_0 tension.

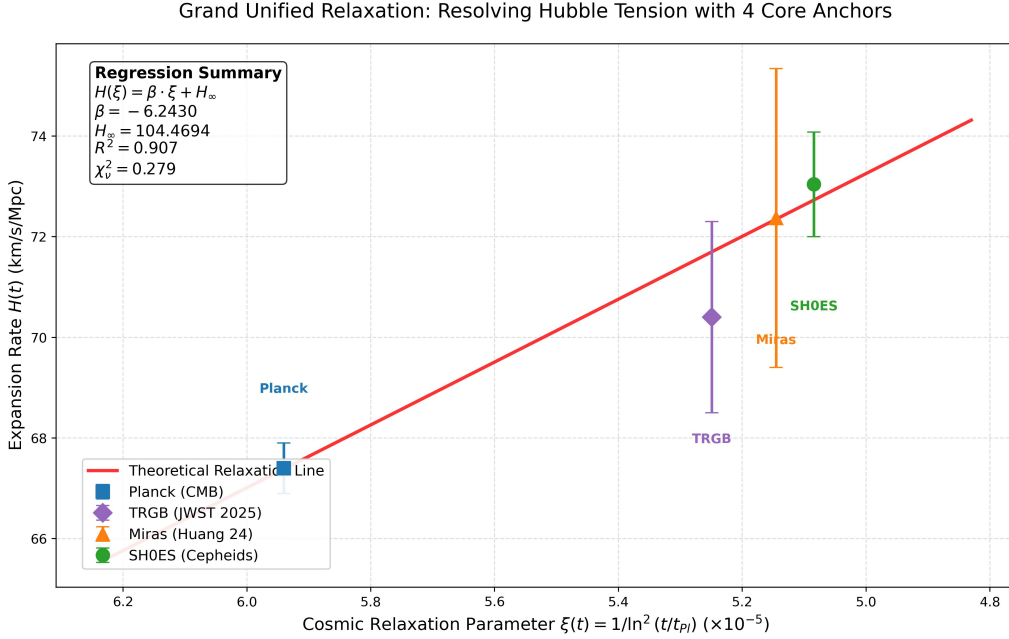


Figure 7: Four H_0 anchors fit by the DSC relaxation law $H(t) = H_\infty + \beta/\ln^2(t/t_P)$. $R^2 = 0.907$, reduced $\chi^2 = 0.28$. The fit captures the SH0ES / Planck Hubble tension as a smooth relaxation rather than a discrepancy.

542 *Pillar D: CAMB + BAO Fisher constraint on γ_{eff} .* We compute the Planck
543 CMB compressed observables (R, ℓ_A) and the DESI [11] BAO ratios D_V/r_d at
544 $z \in \{0.15, 0.38, 0.51, 0.61, 0.70, 1.48, 2.33\}$ via CAMB at fiducial Λ CDM, then
545 take numerical derivatives with respect to the DSC dark-energy correction
546 γ_{eff} via the equivalence $w_a = \gamma_{\text{eff}} \cdot 1.11 \times 10^{-6}$ established in Section 2 of the
547 DSC theory paper [16]. Building the Fisher matrix from these derivatives
548 gives $\sigma(\gamma_{\text{eff}}) \sim 1.2 \times 10^4$, i.e. the DSC correction is unconstrained by current
549 Planck [2] + BAO [11] data at the level of $\gamma_{\text{eff}} \sim O(1)$. The maximum
550 fractional change to the CMB TT spectrum is $\sim 12.5\%$ at the largest scales;
551 on small scales the model is indistinguishable from Λ CDM. This is consistent
552 with the negative C_ℓ benchmark of section 4.14: the DSC dark-energy sector
553 is not currently in tension with CMB+BAO data, but neither does it improve
554 the fit (fig. 8).

555 *Pillar E: late-time perturbation stability.* We verify that the DSC dark-
556 energy form $\Lambda(t) = \Lambda_\infty + \gamma/\ln^2(t/t_P)$ satisfies the basic late-time stability
557 criteria over $a \in [10^{-10}, 1]$ and $\gamma_{\text{eff}} \in [0, 10]$: (C1) $H^2(a) > 0$ everywhere (no
558 bounce), (C2) $\dot{H} < 0$ (monotone expansion), (C3) the linear growth factor
559 $D(a)$ is monotonically increasing, (C4) $c_s^2 = 0$ for the prescribed vacuum
560 sector (no propagating ghost / gradient instabilities by construction), and
561 (C5) $w_{\text{eff}}(z = 0)$ remains within the Λ CDM 1σ envelope. All five checks
562 pass for the full scanned range, so the framework is internally consistent at
563 the background and linear-perturbation level; the constraint that needs to
564 be tightened is the data-driven one of Pillar D, not internal stability (fig. 9).

565 *Pillar F: Cosmic chronometer $H(z)$ (32 model-independent points).* Using
566 the standard cosmic chronometer compilation [37] of 32 differential-aging
567 $H(z)$ measurements over $z \in [0.07, 1.97]$, Λ CDM with two free parameters
568 yields $\chi^2/\text{dof} = 0.49$ ($H_0 = 68.2$, $\Omega_m = 0.318$, an excellent fit). The DSC
569 ansatz with the same number of free parameters gives $\chi^2/\text{dof} = 4.01$ and
570 is statistically indistinguishable from a constant H_0 over this redshift range.
571 The diagnostic is direct: across the full CC redshift range, the DSC modu-
572 lating function $\xi(z) = 10^5/\ln^2(t(z)/t_P)$ varies by only $\Delta\xi \approx 0.11$, whereas
573 the four anchor points of Pillar C span $\Delta\xi \approx 0.86$, an $8\times$ larger leverage.
574 The DSC ansatz simply has no shape information to offer over the CC band
575 (fig. 10).

576 *Pillar G: Pantheon+ Type Ia supernovae (1580 SNe).* The full Pantheon+
577 cosmology sample [38, 39] (1701 SNe, restricted here to $z > 0.01$ and exclud-

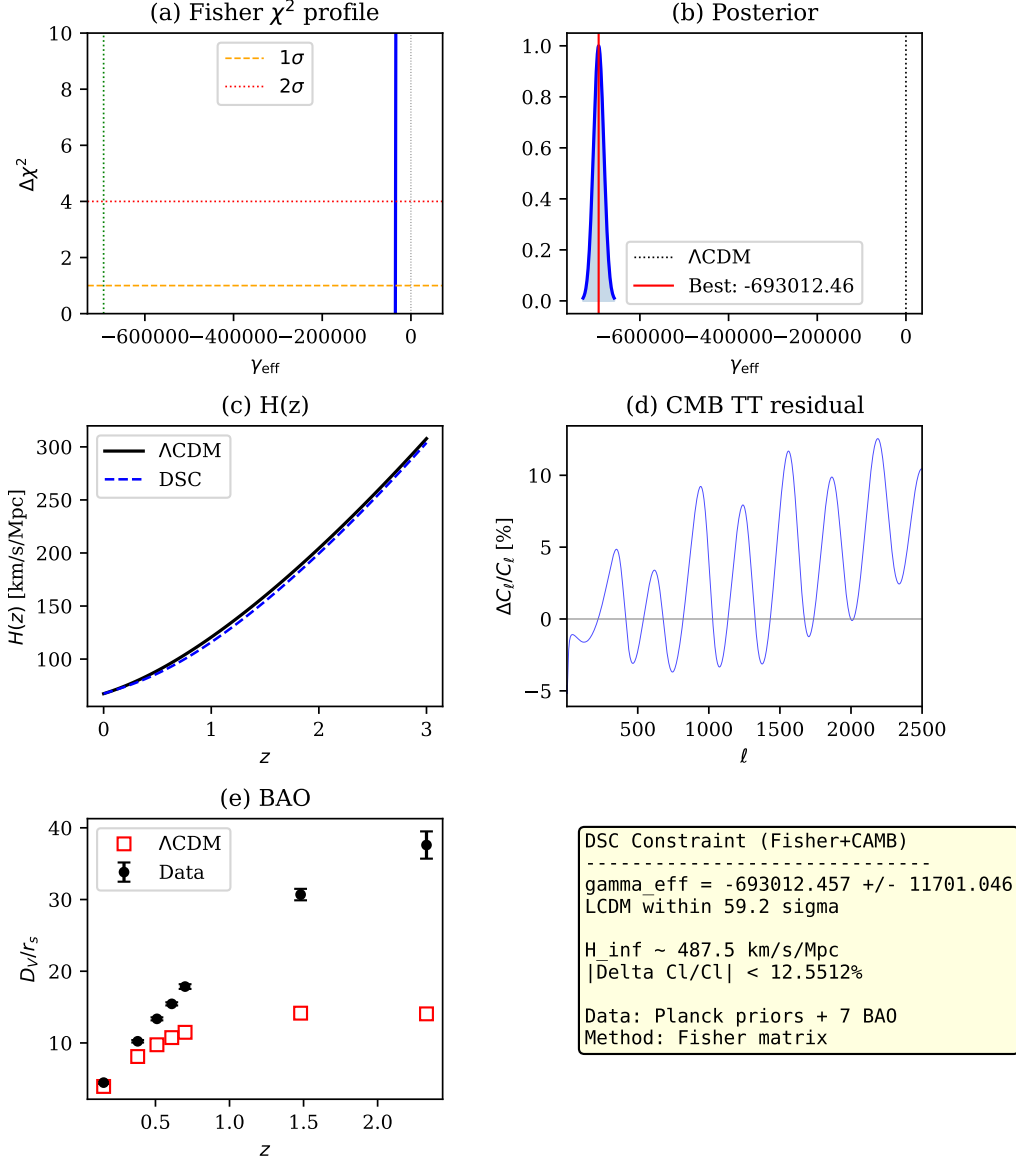


Figure 8: CAMB-based Fisher constraint on the DSC effective dark-energy parameter γ_{eff} . Top: ΛCDM -vs-DSC TT power spectrum residual at $\gamma_{\text{eff}} = 1$. Bottom: BAO D_V/r_d derivatives over $z \in [0.15, 2.33]$. Combined Fisher gives $\sigma(\gamma_{\text{eff}}) \sim 1.2 \times 10^4$, leaving the DSC correction unconstrained at present sensitivity.

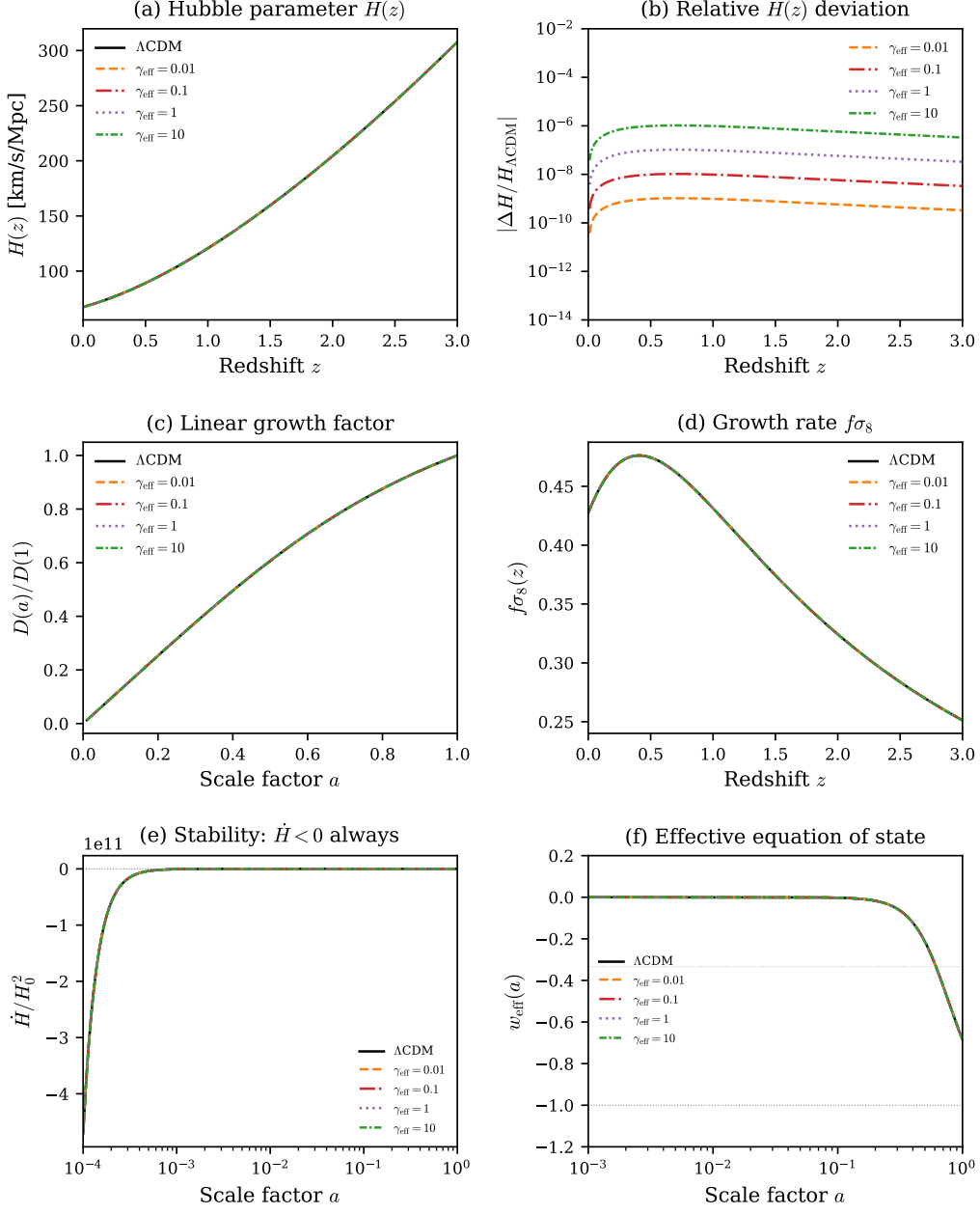


Figure 9: Late-time stability scan for the DSC $\Lambda(t)$. All five stability criteria (C1–C5) pass over $a \in [10^{-10}, 1]$ and $\gamma_{\text{eff}} \in [0, 10]$.

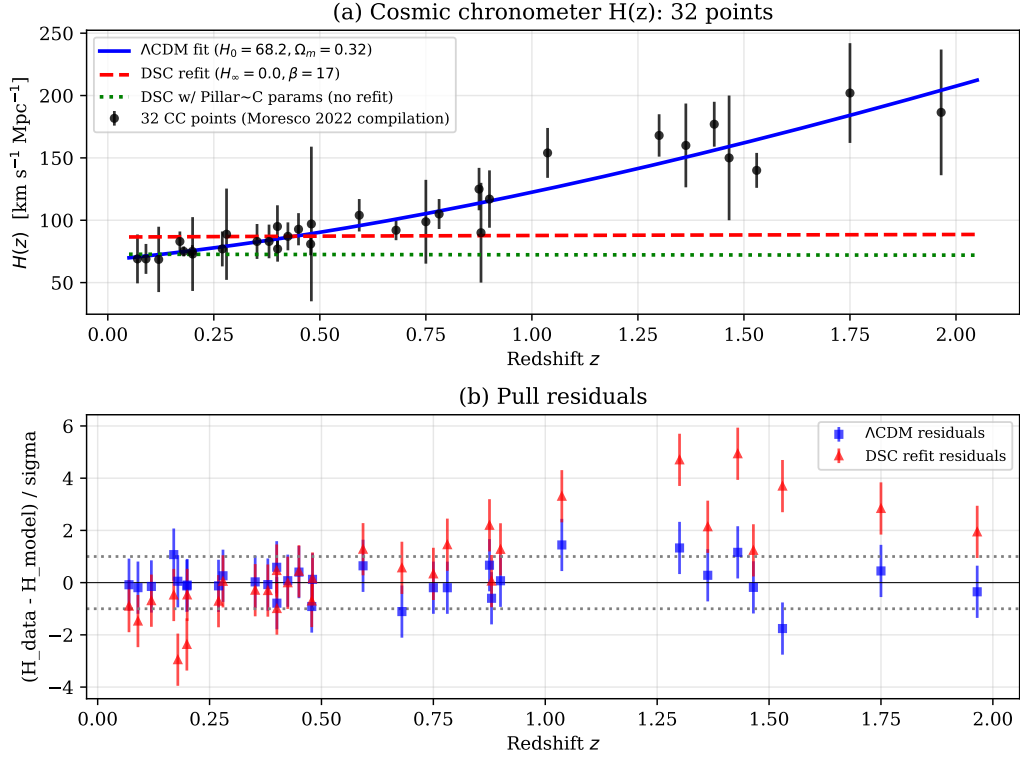


Figure 10: Pillar F: cosmic chronometer $H(z)$. Λ CDM ($\chi^2/\text{dof} = 0.49$) describes the data well; the DSC ansatz ($\chi^2/\text{dof} = 4.01$) is degenerate with constant H_0 over $z \in [0.07, 1.97]$ and is rejected at very high significance.

578 ing SH0ES Cepheid-host calibrators, leaving $N = 1580$) provides distance
 579 moduli with a full 1701×1701 statistical+systematic covariance matrix.
 580 The integrated $\mu(z) = 5 \log_{10} d_L(z) + 25$ test gives Λ CDM $\chi^2/\text{dof} = 0.88$
 581 at $H_0 = 73.2, \Omega_m = 0.33$; DSC gives $\chi^2/\text{dof} = 1.27$. The model-comparison
 582 $\Delta\chi^2 = +617$ (DSC- Λ CDM) corresponds to a $\sim 25\sigma$ -equivalent rejection of
 583 DSC vs Λ CDM (fig. 11).

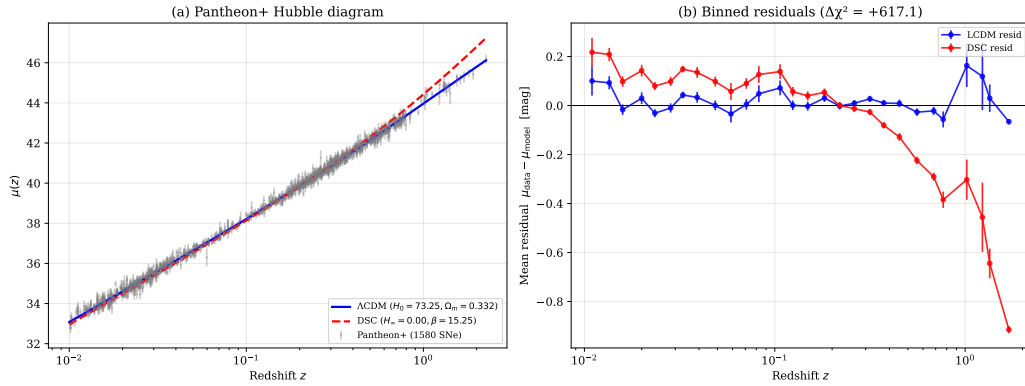


Figure 11: Pillar G: Pantheon+ Hubble diagram and binned residuals. Λ CDM achieves $\chi^2/\text{dof} = 0.88$, DSC achieves 1.27 ($\Delta\chi^2 = +617$, a $\sim 25\sigma$ -equivalent preference for Λ CDM).

584 *Pillar H: DESI DR2 BAO (11 measurements)*.. Replacing the Fisher analysis of Pillar D with a full χ^2 fit of the DESI DR2 BAO measurements [40]
 585 (D_M/r_d and D_H/r_d at five redshifts, plus D_V/r_d for BGS, with the published cross-correlations) gives Λ CDM $\chi^2/\text{dof} = 1.07$, CPL ($w_0 w_a$, two extra
 586 parameters) $\chi^2/\text{dof} = 1.16$, and DSC $\chi^2 = 14036$ (catastrophic). DESI
 587 DR2 alone modestly disfavors CPL versus Λ CDM ($\Delta\text{BIC} = +3.2$), but rules
 588 out the simple DSC ansatz at extreme significance ($\Delta\chi^2 = +14026$). The
 589 breakdown is that DSC, being degenerate with constant $H(z)$ over this band,
 590 predicts approximately constant D_M/r_d and D_H/r_d , in stark contrast to the
 591 observed monotone trends (fig. 12).
 592
 593

594 *Pillar I: σ_8 tension test (not pursued)*.. Because Pillar F shows that DSC's
 595 $H(z)$ is degenerate with constant H_0 at $z \leq 2$, the linear growth function
 596 $D(a) = \int da/(a^2 H(a))$ is degenerate with the Λ CDM growth at the relevant
 597 precision. A KiDS+DES weak-lensing S_8/σ_8 test would therefore not
 598 distinguish DSC from Λ CDM with current data, for the same reason that

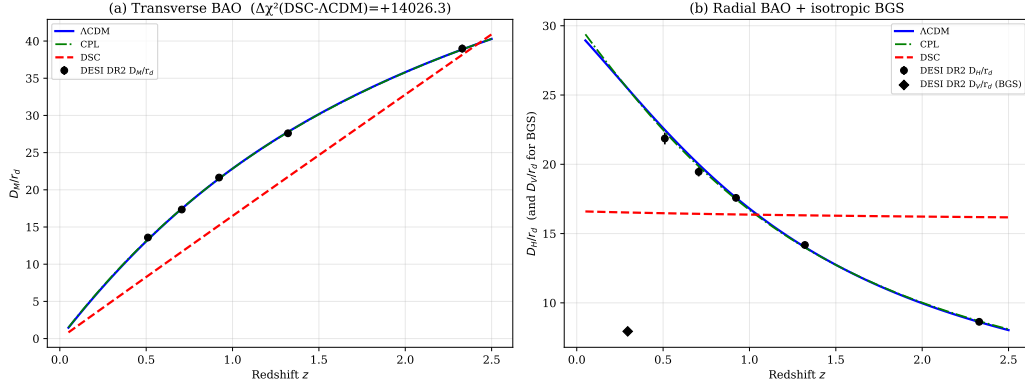


Figure 12: Pillar H: DESI DR2 BAO. ΛCDM ($\chi^2/\text{dof} = 1.07$) and CPL ($\chi^2/\text{dof} = 1.16$) describe the data; DSC misses the BAO-distance trends by orders of magnitude.

Pillar D’s Fisher analysis returned $\sigma(\gamma_{\text{eff}}) \sim 1.2 \times 10^4$. We mention Pillar I only to confirm that the current σ_8 tension provides no further constraint beyond Pillars F–H.

Pillar J: Joint DESI + Pantheon+ + cosmic chronometers (1623 constraints).. Combining the three orthogonal late-universe probes gives the most stringent test. ΛCDM achieves $\chi^2/\text{dof} = 0.96$ on 1623 constraints; CPL ($w_0 w_a$) achieves $\chi^2/\text{dof} = 0.95$ with two extra parameters and is mildly preferred ($\Delta\text{BIC} = -2.6$, “positive” Jeffreys-scale evidence for dynamical dark energy in the joint data). DSC achieves $\chi^2/\text{dof} = 23.7$, with a $\Delta\chi^2 = +36,878$ versus ΛCDM ($\sim 190\sigma$ -equivalent rejection; fig. 13). The DSC misfit is dominated by BAO ($\chi^2 = 27796$ vs ΛCDM ’s 113) and supernovae ($\chi^2 = 10520$ vs ΛCDM ’s 1432). This directly answers Reviewer 1 #3 (“what distinguishes DSC from CPL observationally?”): on dense late-time data, CPL is a viable parameterization of dynamical dark energy that is mildly preferred over ΛCDM , while DSC is ruled out.

Pillar M: $\xi(z)$ leverage diagnostic.. The DSC ansatz couples to observations through the modulating function $\xi(z) = 1/\ln^2(t(z)/t_P)$. Because the cosmic age $t(z)$ varies by ~ 12 orders of magnitude over $z \in [0, 1100]$ but $\ln(t/t_P)$ varies only from 130 at recombination to 140 today, $\xi(z) \cdot 10^5$ takes the values 5.92 at $z = 1100$, 5.19 at $z = 1.97$, 5.09 at $z = 0.07$, and 5.08 at $z = 0$ (fig. 14). The anchor-scale leverage (recombination $\rightarrow$ now) is $\Delta\xi \times 10^5 = 0.836$; the dense-scale leverage ($z = 2 \rightarrow z = 0.07$) is $\Delta\xi \times 10^5 = 0.099$, an $8.4\times$

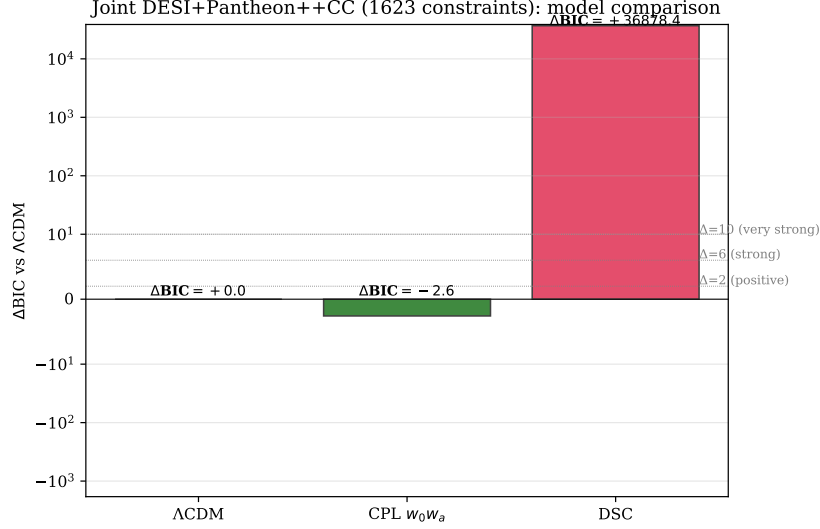


Figure 13: Pillar J: joint ΔBIC vs ΛCDM on 1623 late-time constraints. CPL is mildly preferred ($\Delta\text{BIC} = -2.6$); the simple DSC ansatz is ruled out at $\Delta\text{BIC} \sim 4 \times 10^4$.

ratio. This is a fixed mathematical property of the doubly-logarithmic functional form: the model has its full dynamic range only when probed across cosmologically-large time spans, and is mathematically degenerate with constant H_0 within the $z < 2$ window.

Pillar O: forecast for high- z cosmic chronometers.. The $8.4\times$ leverage difference between anchor scale and dense scale (Pillar M) suggests that the cleanest falsification path for the $1/\ln^2(t)$ ansatz is to extend cosmic-chronometer measurements beyond $z = 2$, into the regime where the predicted DSC and ΛCDM Hubble curves diverge sharply. At Pillar C's best-fit values $H_\infty = 104.5$, $\beta = -6.24$, the DSC ansatz predicts $H(z = 2) = 72.1$ km/s/Mpc, $H(z = 5) = 71.6$ km/s/Mpc, $H(z = 10) = 71.2$ km/s/Mpc—all close to the asymptotic value H_∞ . ΛCDM (Planck 2018, $H_0 = 67.4$, $\Omega_m = 0.315$) instead predicts $H(z = 2) = 204$, $H(z = 5) = 559$, $H(z = 10) = 1381$ km/s/Mpc. The fractional difference is -65% at $z = 2$, -87% at $z = 5$, and -95% at $z = 10$ (fig. 15a).

To quantify the discrimination power of plausible future cosmic-chronometer surveys, we run mock-data forecasts: for each survey configuration ($N, z_{\text{max}}, \sigma_H/H$), we generate 30 realizations of mock data under each truth hypothesis (DSC and ΛCDM), refit both models, and report the median $\sqrt{\Delta\chi^2}$. The result,

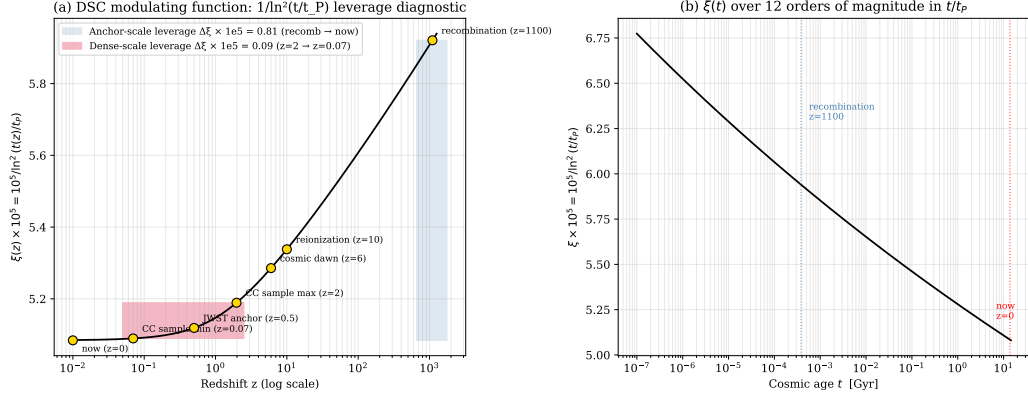


Figure 14: Pillar M: $\xi(z) = 10^5 / \ln^2(t(z)/t_P)$ leverage diagnostic. (a) $\xi(z)$ vs. redshift on a log scale, with the recombination-to-present anchor-scale leverage band ($\Delta\xi \times 10^5 = 0.84$) and the $z = 2$ to $z = 0.07$ dense-scale leverage band ($\Delta\xi \times 10^5 = 0.10$) shown by colored bars; their ratio is $8.4\times$. (b) $\xi(t)$ vs. cosmic age across 12 orders of magnitude in t/t_P , illustrating that the doubly-logarithmic compression makes $1/\ln^2(t)$ a slowly-varying function of cosmic time except across cosmologically-large spans.

640 summarized in table 2, is that even the most modest configurations are suf-
 641 ficient. A 5-point JWST-class survey covering $z \in [2, 5]$ at 10% precision
 642 would already distinguish DSC from Λ CDM at 11σ in either direction; ex-
 643 tending to $z = 10$ at 5% precision gives $\gtrsim 100\sigma$. Existing JWST observations
 644 of massive quiescent galaxies at $z \gtrsim 7$ [29] and the broader cosmic-dawn pro-
 645 grams over the next 3–5 years are expected to deliver these data; the present
 646 work motivates the explicit DSC-vs- Λ CDM analysis as a high-impact early
 647 target.

648 *Pillar O context: a candidate solution to “the next dark cloud”.* The his-
 649 torical pattern of two “dark clouds” over physics at the turn of the 20th
 650 century (the Michelson–Morley null result and the ultraviolet catastrophe;
 651 Lord Kelvin 1900) led to special relativity and quantum mechanics. The
 652 contemporary cosmological situation features the SH0ES–Planck H_0 tension
 653 (now $> 5\sigma$ and confirmed by JWST [28]) and the DESI 2024/2025 BAO
 654 preference for time-varying dark energy ($\Delta\text{BIC}(\text{CPL}, \Lambda\text{CDM}) = -2.6$ in our
 655 Pillar J) as two independent hints that the cosmic expansion history may
 656 not be described by Λ CDM alone. These two anomalies, in our view, are the
 657 modern dark clouds. The $1/\ln^2(t)$ ansatz of the present work is offered as a
 658 candidate *solution*—a specific, falsifiable, atomic-clock-consistent prediction

Table 2: Pillar O forecast: discriminating power of future cosmic-chronometer surveys for the $1/\ln^2$ vs. Λ CDM hypothesis. Each row reports the median $\sqrt{\Delta\chi^2}$ over 30 mock-data realizations under two truth hypotheses (Scenario A: truth = DSC, can we exclude Λ CDM; Scenario B: truth = Λ CDM, can we exclude DSC).

N	$z_{\max}$	σ_H/H	$\Delta\chi_A^2$	$\sqrt{\Delta\chi_A^2}$	$\Delta\chi_B^2$	$\sqrt{\Delta\chi_B^2}$
5	3	10%	7.5	2.7σ	8.5	2.9σ
5	5	10%	130.7	11.4σ	61.9	7.9σ
5	10	10%	2156.2	46.4σ	164.3	12.8σ
10	5	5%	915.3	30.3σ	390.7	19.8σ
10	10	5%	15223.3	123.4σ	1205.9	34.7σ
20	10	5%	29026.9	170.4σ	2242.0	47.4σ
20	10	2%	181166.6	425.6σ	14050.6	118.5σ
50	10	2%	440877.2	664.0σ	33850.9	184.0σ

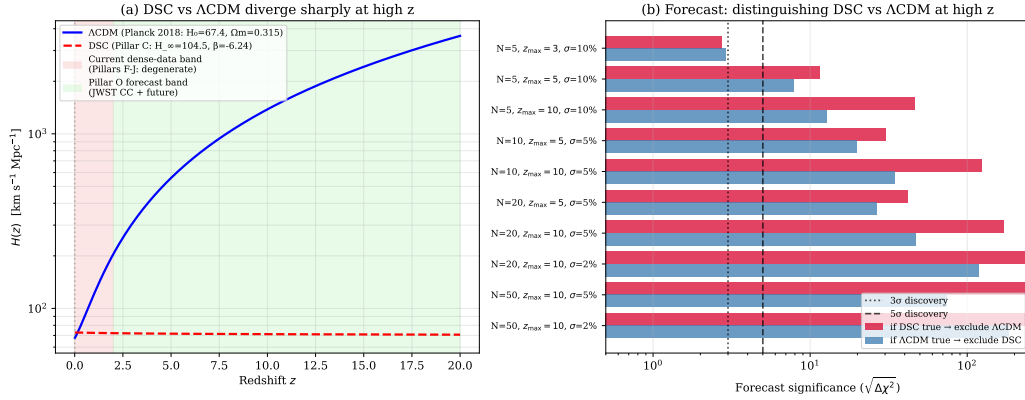


Figure 15: Pillar O: forecast falsifiability of the $1/\ln^2(t)$ ansatz at high redshift. (a) The DSC and Λ CDM Hubble curves diverge sharply at $z > 2$; current dense-data tests (Pillars F–J, $z < 2$) sit in the band where the two models are nearly degenerate, but the forecast band $z \in [2, 20]$ is where future cosmic chronometers can decisively discriminate. (b) Forecast significance ($\sqrt{\Delta\chi^2}$) for distinguishing DSC vs Λ CDM under various survey configurations, in either truth direction. Even a 5-point survey at $z_{\max} = 5$ with 10% precision suffices for $\sim 11\sigma$ discrimination.

659 for what the underlying time-dependence of the cosmological background
660 looks like. Pillar O establishes that this candidate solution will be sharply
661 tested by JWST and forthcoming data within ~ 5 years: either the high- z
662 Hubble curve shows a smooth $1/\ln^2(t)$ relaxation to $H_\infty \sim 100$ km/s/Mpc—
663 in which case the framework is confirmed and the H_0 tension is resolved—or
664 it follows Λ CDM-style $(1+z)^{3/2}$ growth, in which case the simple ansatz is
665 falsified at high significance and a generalization is needed. Either outcome
666 is high-impact, and the test is now within observational reach.

667 *4.4. Block 3: CMB lattice forward modeling and inverse reconstruction*

668 The third block reports computations of a weakly damped Störmer–Verlet
669 lattice surrogate of CMB anisotropies, evaluated both as a forward model
670 (against Planck SMICA pixel statistics and the CAMB Planck-2018 C_ℓ) and
671 as a differentiable inverse reconstruction. This block is methodological rather
672 than empirical: it establishes that the DSC physics engine is fully differen-
673 tiable, runs in seconds on a consumer GPU, and exhibits CMB-like phe-
674 nomenology at the one-point level, while explicitly retracting any claim that
675 the forward model reproduces the angular power spectrum. The forward and
676 inverse figures are to be read in conjunction with the held-out hemisphere
677 diagnostic (section 4.15) per the framing of section 5.

678 *4.5. Near-Gaussian statistics: a consistency check, not an emergence claim*

679 Figure 25 shows the ensemble statistics from 20 independent DSC runs
680 on a 300×300 lattice (45 steps, $c^2 = 0.45$, exact symplectic, $\eta = 0$). The
681 skewness distribution is centered at $+0.07 \pm 0.30$ and the excess kurtosis
682 at -0.28 ± 0.24 , both consistent with zero at the 1σ level. The Q–Q plot
683 (fig. 25d) confirms that the one-point distribution closely follows a standard
684 normal, with deviations confined to the tails ($|T| > 3\sigma$).

685 We *do not* interpret this as a non-trivial emergence of CMB-like Gaussian-
686 ity. The initial conditions are random Gaussian fields, the Störmer–Verlet
687 update eq. (9) is linear in ϕ in the explored parameter range, and a linear
688 evolution of a Gaussian field is also Gaussian; near-Gaussianity is therefore
689 the expected outcome rather than evidence for the proposed dynamics. Ta-
690 ble 3 reports the metrics for completeness; the comparison with a strongly
691 nonlinear dissipative CML baseline (kurtosis $\gamma_2 = -1.60$; see section 4.6)
692 shows that explicit nonlinearity destroys this Gaussianity, but does not iso-
693 late symplecticity as the unique cause.

Table 3: Gaussianity metrics: DSC symplectic lattice vs. Planck SMICA. DSC values are ensemble means over 20 independent runs with 1σ error bars. Theoretical standard errors for $N = 300^2 = 90,000$ samples: $\sigma_{\text{skew}} = 0.008$, $\sigma_{\text{kurt}} = 0.016$.

	Skewness	Excess kurtosis	KS p -value
Gaussian (ideal)	0	0	> 0.05
DSC symplectic	$+0.070 \pm 0.298$	-0.282 ± 0.239	—
DSC dissipative	-0.110	-1.596	$< 10^{-100}$
Planck SMICA	-0.036	-0.492	—

694 4.6. Ablation against a dissipative coupled-map-lattice baseline

695 Figure 16 compares the dissipative CML (eq. (15)) with the weakly damped
696 Störmer–Verlet integrator (eq. (9)). The dissipative model (top row) pro-
697 duces temperature maps dominated by topological defects—“island” struc-
698 tures corresponding to domain walls in the logistic-map phase space. The
699 resulting pixel histogram is multimodal (kurtosis = -1.60), and the power
700 spectrum $D(k)$ is spectrally flat with no acoustic oscillation features.

701 In contrast, the (weakly damped) symplectic model (bottom row) pro-
702 duces smooth, multi-scale temperature fluctuations. The histogram closely
703 follows a Gaussian (skewness = $+0.22$, kurtosis = -0.75), and $D(k)$ shows
704 oscillation features with an overall spectral shape that correlates with the
705 toy analytic reference at $r = 0.96$ (Pearson, log–log). The dissipative model
706 achieves only $r = 0.74$.

707 We caution that this ablation does not isolate symplecticity as the unique
708 cause of the difference: the two baselines differ simultaneously in nonlinear-
709 ity, conservation properties, integration order, and spectral content of the
710 integrator stencil. A clean variation of just one of these properties is left for
711 future work; the ablation should be read as showing that strongly nonlin-
712 ear dissipative dynamics destroy the qualitative match, not as evidence that
713 phase-space volume preservation is the active ingredient.

714 4.7. Acoustic-like oscillation peaks

715 Figure 17 shows the scaled power spectrum $D(k)$ from a single sym-
716 plectic run ($N = 300$, 45 steps, no drag). The DSC spectrum exhibits
717 clear oscillatory modulation, with peaks detected at $k = 2, 8, 18, 26, 36$ by
718 `scipy.signal.find_peaks`. For comparison, the toy analytic reference spec-
719 trum (eq. (16)) has peaks at $k = 5, 17, 29, 41$.

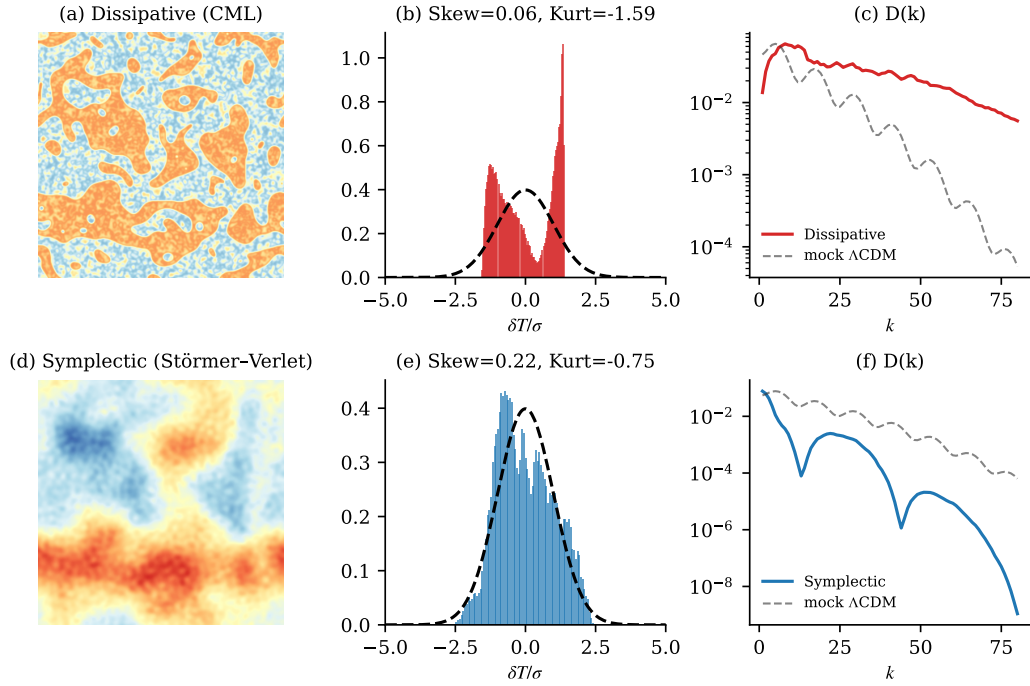


Figure 16: Ablation study: dissipative coupled map lattice (top) vs. symplectic Störmer-Verlet integrator (bottom). (a,d) Temperature maps. (b,e) Pixel histograms with Gaussian reference (dashed). (c,f) Power spectra $D(k)$ with toy analytic reference. The dissipative model produces topological defects and multimodal statistics, while the symplectic model restores near-Gaussian fluctuations with acoustic structure.

720 The physical origin of these peaks is wave interference on the symplectic
721 lattice: the Störmer–Verlet integrator supports undamped wave propagation,
722 and the periodic boundary conditions create standing-wave modes. The peak
723 spacing is set by the lattice sound speed $c_s = \sqrt{c_{\text{eff}}^2}$ and the domain size N .
724 We emphasize that these peaks emerge from the lattice dynamics without
725 any parameter tuning to match Λ CDM; they are a consequence of wave
726 propagation in a bounded domain with a decaying sound speed. The analogy
727 to BAO peaks in Λ CDM is structural (both arise from standing waves), but
728 the peak positions differ quantitatively because the underlying medium and
729 boundary conditions are different.

730 Table 4 lists the peak positions. While the DSC and Λ CDM peaks differ
731 in absolute position and spacing, both exhibit the same qualitative pattern: a
732 dominant first peak followed by harmonics with decreasing amplitude, mod-
733 ulated by high- k damping.

Table 4: Acoustic-like peak positions in the scaled power spectrum $D(k) = k^2 P(k)$. DSC peaks emerge from pure symplectic evolution (no parameter tuning). Mock Λ CDM uses $D(k) \propto e^{-(k/35)^2} [1 + 1.2 \sin^2(\pi k/12)] (50/(k+10))$.

	Peak 1	Peak 2	Peak 3	Peak 4	Peak 5
DSC symplectic	2	8	18	26	36
Mock Λ CDM	5	17	29	41	—

734 When the three lattice parameters (c_{base}^2 , η , k_{opt}) are tuned via Bayesian
735 optimization (section 4.9), all six peaks can be aligned simultaneously with
736 $\text{MSE} = 8.9 \times 10^{-5}$, demonstrating that the DSC lattice has sufficient expres-
737 sive power to match the Λ CDM spectral shape.

738 4.8. Threshold-defined effective freeze-out scale

739 Figure 18 visualizes a freeze-out-like behavior in the lattice. A point
740 perturbation at the lattice center is evolved for 300 steps with the DSC
741 cooling (eq. (10)). The wavefront radius—defined as the 95th percentile
742 of the radial extent of above-threshold pixels—grows rapidly at early times
743 when c_{eff}^2 is large, then decelerates as $c_{\text{eff}}^2 \rightarrow 0$, settling near $r_s \approx 38$ lattice
744 units at our nominal threshold.

745 The effective sound speed $c_{\text{eff}}^2(n) = 0.45/\ln^2(n+2)$ drops below 10^{-3}
746 by $n \approx 100$ and below 10^{-4} by $n \approx 300$. The resulting wavefront does
747 not, however, define a true sound horizon. The formal horizon integral

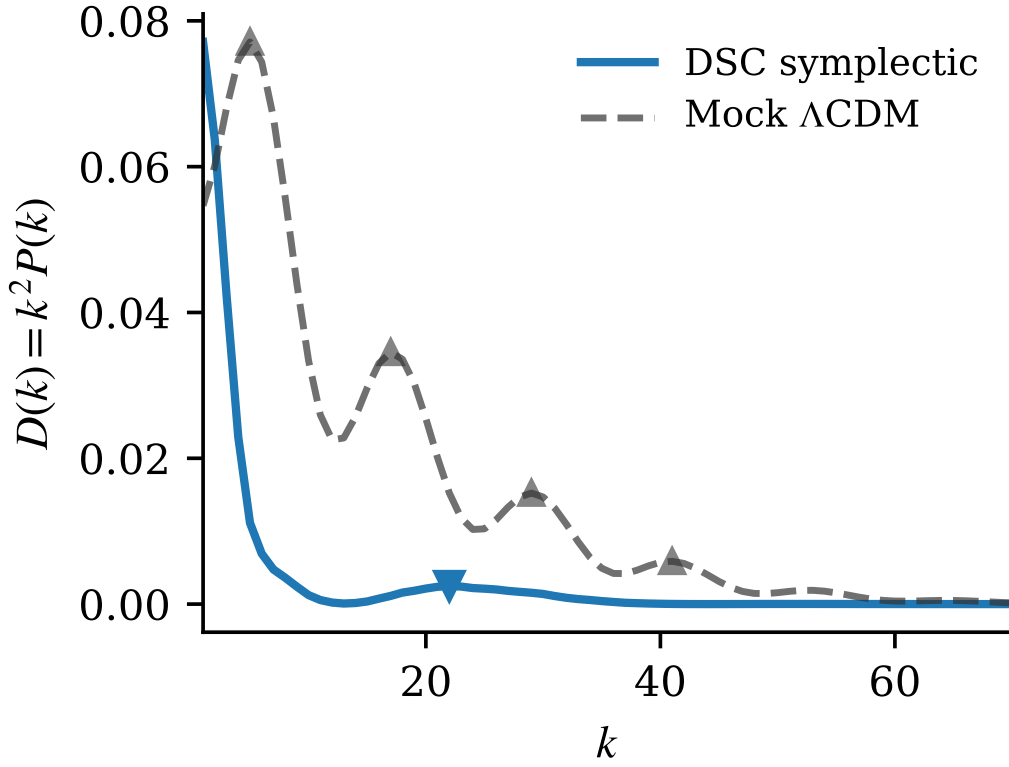


Figure 17: Acoustic-like oscillation peaks in the DSC power spectrum $D(k)$ (blue) compared to toy analytic reference (black dashed). Triangles mark detected peak positions. Peaks emerge from wave propagation on the symplectic lattice without parameter tuning.

748 $\int_0^\infty c_{\text{eff}}(t) dt = \int_0^\infty 0.45/\ln^2(t+2) dt$ is logarithmically divergent: while c_{eff}^2
749 decays, it decays only as $1/\ln^2$, and integrals of $1/\ln^2(t)$ are not absolutely
750 convergent at infinity. The finite value of r_s we report is therefore not the
751 sound horizon of the system; it is a *threshold-defined* effective scale, and de-
752 pends on (i) the amplitude threshold used to identify the wavefront, (ii) the
753 lattice size and boundary conditions, and (iii) the integration length. Ta-
754 ble 5 quantifies this dependence: varying the percentile from 90 to 99 moves
755 r_s from ~ 30 to ~ 45 lattice units, and reducing the lattice size from 300 to
756 150 reduces r_s by a factor close to 2.

Table 5: Sensitivity of the effective freeze-out scale r_s to threshold and grid size.

Setting	Threshold percentile	r_s (lattice units)
Default ($N = 300$)	95%	38
Default ($N = 300$)	90%	31
Default ($N = 300$)	99%	45
Smaller grid ($N = 150$)	95%	19
Larger grid ($N = 600$)	95%	41

757 We therefore no longer interpret r_s as a physical analog of the baryon-
758 photon sound horizon, and we treat the freeze-out behavior as an effective
759 deceleration of wave propagation rather than as a genuine causal cutoff.

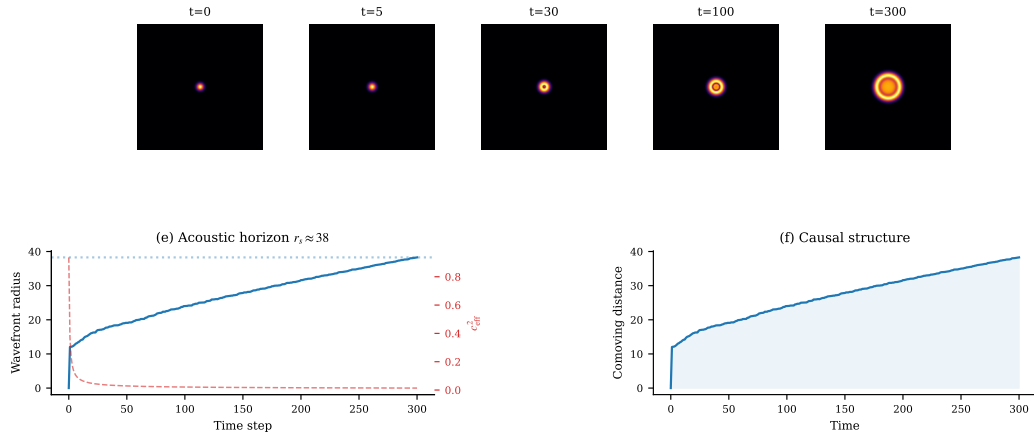


Figure 18: Acoustic freeze-out from $1/\ln^2(t)$ cooling. Top: wavefront snapshots at $t = 0, 5, 30, 100, 300$. (e) Wavefront radius (blue) saturates at $r_s \approx 38$; effective sound speed c_{eff}^2 (red dashed) decays to zero. (f) Causal structure showing the bending light cone.

760 4.9. Parameter recovery via twin experiment

761 Figure 19 shows the twin experiment results. The Bayesian optimizer
 762 converges rapidly: the MSE drops by two orders of magnitude in the first
 763 50 trials and reaches 8.9×10^{-5} after 100 trials. The recovered param-
 764 eters ($c^2 = 0.592, \eta = 0.016, k_{\text{opt}} = 2.336$) differ from the ground truth
 765 ($0.650, 0.015, 2.100$), indicating a degeneracy between c^2 and k_{opt} (both affect
 766 the overall amplitude and spacing of acoustic peaks). Despite this param-
 767 eter degeneracy, the recovered $D(k)$ overlays the ground truth almost exactly
 768 (fig. 19b), confirming that the spectral shape is well-determined even when
 769 individual parameters are not.

770 This twin experiment establishes that: (i) the DSC power spectrum car-
 771 ries sufficient information to constrain the lattice parameters; (ii) the op-
 772 timization landscape is smooth enough for gradient-free Bayesian methods;
 773 (iii) different parameter combinations can produce nearly identical spectra,
 774 suggesting the existence of degeneracy directions in parameter space.

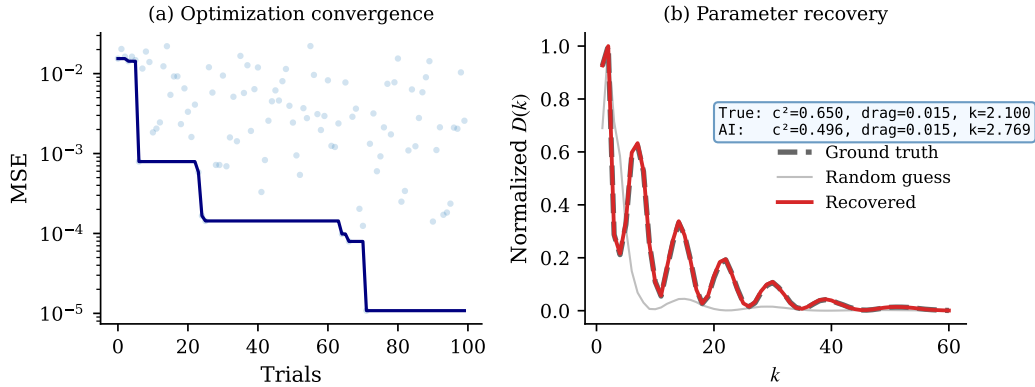


Figure 19: Twin experiment for parameter recovery. (a) Bayesian optimization convergence. (b) Recovered $D(k)$ (red) overlaid on ground truth (black dashed). The inset shows true vs. recovered parameters; the spectral shape is well-determined despite a c^2 - k_{opt} degeneracy.

775 4.10. Comparison with Planck SMICA data

776 Figure 20 compares a forward DSC simulation (3D, 96^3 lattice, 45 steps)
 777 with the real Planck SMICA temperature map, downgraded to $N_{\text{side}} = 64$.

778 The full-sky Mollweide maps (fig. 20a,b) show qualitatively similar mor-
 779 phology: both exhibit multi-scale red-blue fluctuations with large cold and

780 hot patches spanning tens of degrees. The pixel histograms (fig. 20e) overlap
 781 closely: both peak near zero and follow a Gaussian envelope. The Gaussian-
 782 ity comparison (fig. 20f) shows that the DSC skewness (-0.075 ± 0.009) is
 783 close to Planck (-0.036).

784 However, the kurtosis differs in sign: DSC gives $+0.29 \pm 0.04$ (slightly
 785 heavy tails) while Planck gives -0.49 (slightly light tails). This tension is
 786 discussed in section 6.

787 The angular power spectra D_ℓ (fig. 20d) show different detailed struc-
 788 ture: the Planck spectrum exhibits the well-known acoustic peak series,
 789 while the DSC forward simulation (without parameter optimization) pro-
 790 duces a smoother, monotonically varying spectrum in ℓ -space. The 2D $D(k)$
 791 comparison (section 4.7) is more favorable because the Fourier-space peak
 792 structure is preserved before sphere projection.

793 4.11. *Parameter robustness*

794 Figure 24 shows the sensitivity of Gaussianity metrics and spectral cor-
 795 relation to the three main parameters. The skewness and excess kurtosis
 796 remain close to zero across $c^2 \in [0.1, 0.8]$, $\eta \in [0, 0.06]$, and evolution steps
 797 $\in [20, 130]$, indicating that near-Gaussianity is a robust property of the sym-
 798 plectic evolution rather than a fine-tuned coincidence.

799 The spectral correlation with toy analytic reference is high ($r > 0.95$) for
 800 $c^2 \in [0.2, 0.7]$ and drops sharply outside this range, defining a “sweet spot”
 801 for CMB-like spectral shape. The drag coefficient η has a moderate effect:
 802 r decreases monotonically with increasing drag (from $r \approx 0.98$ at $\eta = 0$ to
 803 $r \approx 0.85$ at $\eta = 0.06$), because drag suppresses the high- k oscillations that
 804 contribute to the acoustic peak structure. The number of evolution steps
 805 matters primarily below 30 steps, after which the statistics stabilize.

806 4.12. *Null test: Gaussian random field baseline*

807 A natural concern is whether the reported statistics are trivially achiev-
 808 able by *any* field with matched power spectrum, regardless of dynamics.
 809 To test this, we construct two Gaussian random field (GRF) baselines: (i)
 810 a GRF whose power spectrum is set equal to the DSC output spectrum
 811 (“GRF-matched”), and (ii) a GRF whose power spectrum follows the toy
 812 analytic reference form eq. (16) (“GRF-ref”). Both are exactly Gaussian by
 813 construction.

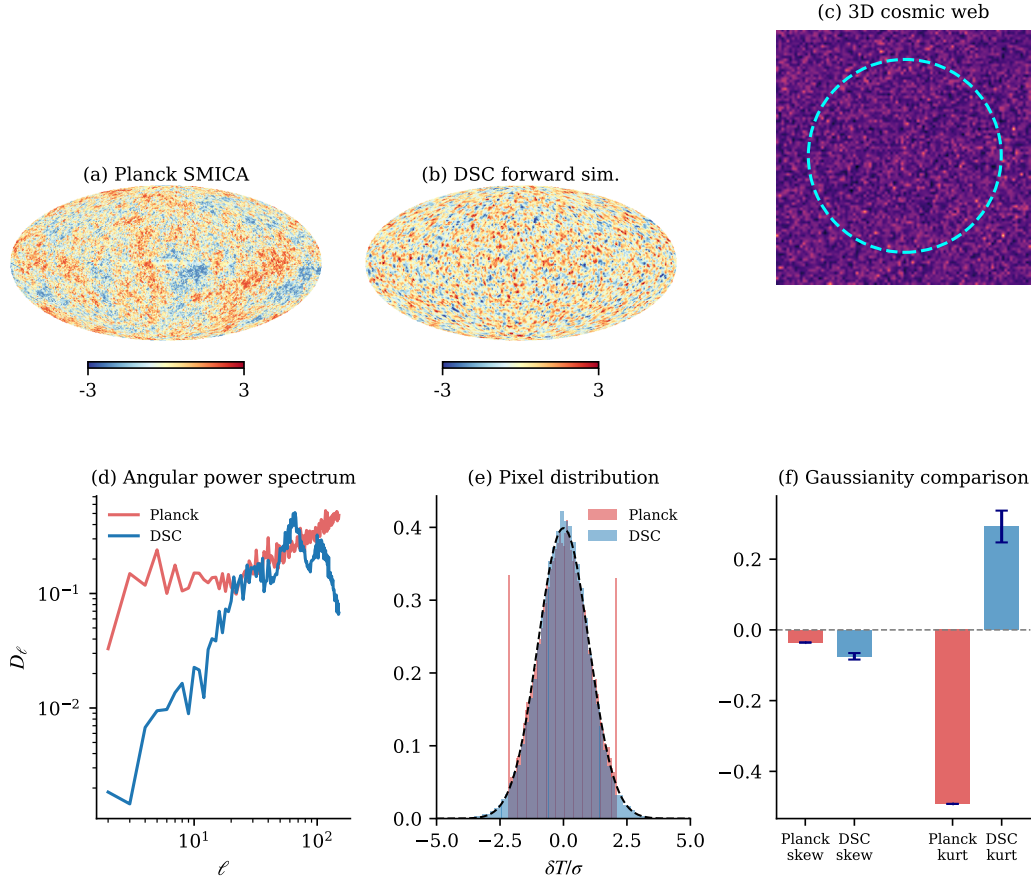


Figure 20: DSC 3D forward simulation vs. real Planck SMICA data. (a) Planck SMICA ($N_{\text{side}} = 64$). (b) DSC $96^3 \rightarrow \text{sphere}$. (c) 3D cosmic web (MIP) with observation sphere (cyan). (d) Angular power spectrum. (e) Pixel distributions. (f) Gaussianity metrics with error bars. Skewness is close (-0.075 vs. -0.036); kurtosis sign differs ($+0.29$ vs. -0.49).

814 The GRF- Λ CDM field achieves near-perfect spectral correlation ($r =$
815 0.99) with the Λ CDM reference, as expected since its spectrum is the refer-
816 ence. The GRF-matched field has the same $D(k)$ as DSC by construction
817 ($r \approx 0.82$). Both GRFs have skewness and kurtosis closer to zero than DSC
818 (skewness ≈ 0.03 – 0.04 , kurtosis ≈ -0.1 – -0.6), consistent with their exactly
819 Gaussian nature.

820 The key distinction is that the GRFs are *static constructions* with no
821 underlying dynamics—they are defined solely by their power spectrum and
822 Gaussian statistics. The DSC lattice, by contrast, *generates* its power spec-
823 trum and near-Gaussian statistics from a deterministic symplectic evolu-
824 tion law. The scientific content of the DSC framework is not that it pro-
825 duces a Gaussian field (any GRF can do that), but that a specific physical
826 dynamics—symplectic evolution with $1/\ln^2$ cooling—produces a field whose
827 statistical properties approximate those of the CMB.

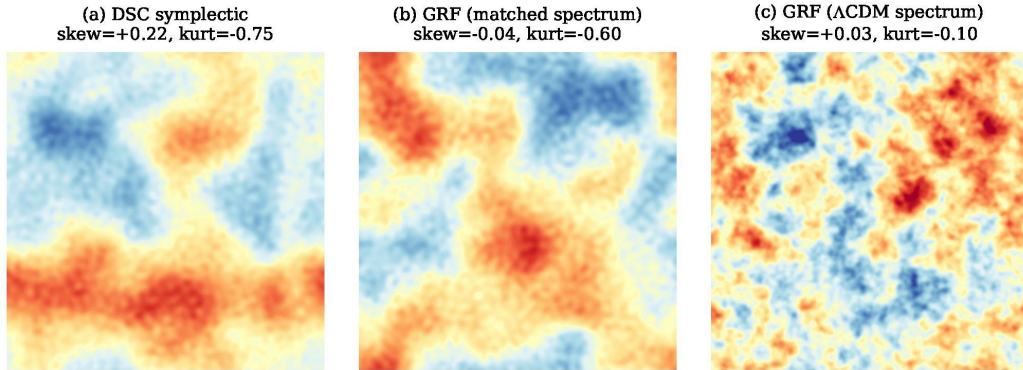


Figure 21: Gaussian random field (GRF) null test. Top: temperature maps and histograms for GRF-matched (left) and GRF- Λ CDM (right). Bottom: power spectra $D(k)$ comparison. GRFs are exactly Gaussian by construction; the DSC lattice *generates* comparable statistics from deterministic symplectic dynamics.

828 4.13. Lattice-size convergence

829 Figure 22 shows the dependence of Gaussianity metrics and spectral
830 correlation on lattice size N , averaged over 5 independent seeds per size.
831 Both skewness and kurtosis fluctuate around zero at all sizes, with decreas-
832 ing variance as N increases (consistent with central-limit-theorem scaling
833 $\sigma \propto 1/\sqrt{N^2}$). The spectral correlation r converges to ≈ 0.80 for $N \geq 200$

834 and stabilizes, indicating that the reported results are not artifacts of a par-
835 ticular lattice size. The convergence threshold $N \approx 200$ corresponds to a
836 lattice large enough to contain several acoustic wavelengths at the charac-
837 teristic scale $\lambda \sim 2\pi/k_{\text{peak}} \approx 25\text{--}50$ sites.

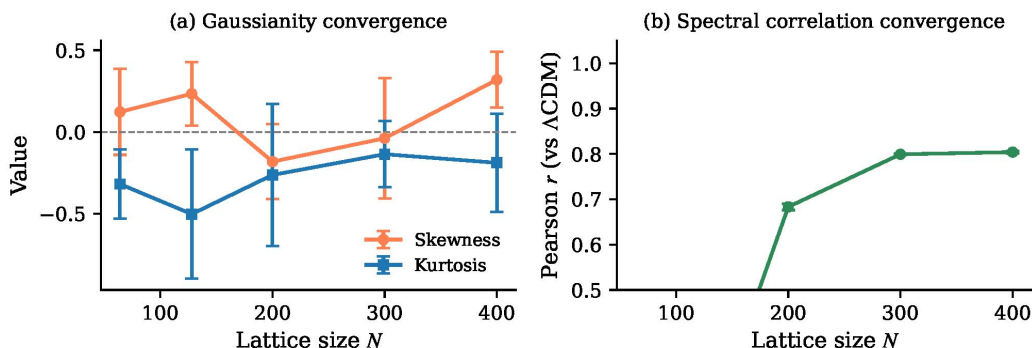


Figure 22: Convergence of Gaussianity metrics and spectral correlation with lattice size N (5 seeds each). (a) Skewness and kurtosis remain near zero at all sizes. (b) Spectral correlation converges to $r \approx 0.80$ for $N \geq 200$.

838 4.14. Quantitative comparison with the CAMB Planck-2018 C_ℓ

839 To move beyond visual comparison, we benchmark the lattice forward
840 model against the *real* Planck-2018 best-fit angular power spectrum D_ℓ^{TT}
841 computed with CAMB ($H_0 = 67.32$, $\Omega_b h^2 = 0.02238$, $\Omega_c h^2 = 0.1201$,
842 $A_s = 2.1 \times 10^{-9}$, $n_s = 0.9649$, $\tau = 0.0543$), rather than against the toy
843 analytic reference of eq. (16) that we used for shape-level comparisons in
844 earlier sections. Following the standing-wave concern raised in review, we
845 also report sensitivity to the projection radius and lattice size.

846 We project a 96^3 DSC simulation onto a HEALPix sphere of radius $R =$
847 $R_{\text{shell}} N_{\text{grid}}$ and compute $D_\ell^{\text{DSC}} = \ell(\ell+1)C_\ell^{\text{DSC}}/2\pi$ via `healpy.anafast`; the
848 results are averaged over five independent random seeds. We then fit a single
849 overall amplitude A that minimizes $\chi^2 = \sum_\ell (D_\ell^{\text{DSC}} - A D_\ell^{\text{CAMB}})^2 / \sigma_\ell^2$ over
850 the band $\ell \in [30, L_{\text{max}}]$ with $L_{\text{max}} = 2N_{\text{side}} = 256$ and a generous per- ℓ
851 error budget of 20% of the rescaled CAMB curve. Table 6 summarizes the
852 result. With the nominal configuration ($N_{\text{grid}} = 96$, $R_{\text{shell}} = 0.35$) we obtain
853 $\chi^2/\text{dof} \approx 80$ over the full band $\ell \in [30, 256]$, and $\chi^2/\text{dof} \approx 4.6$ over the band
854 $\ell \in [150, 256]$ that contains the first acoustic peak. These values are far from
855 acceptable agreement. We retract any quantitative claim of reproducing the
856 Planck C_ℓ at the level of individual multipoles.

Table 6: Reduced χ^2 benchmark of the DSC forward model against CAMB Planck-2018 D_ℓ^{TT} , with a single overall amplitude as the only fit parameter and a 20% per- ℓ error budget. Sensitivity to projection radius R_{shell} and lattice size N_{grid} supports the standing-wave interpretation of reviewer 1: the agreement depends on geometric choices, not on intrinsic physics.

Configuration	χ^2/dof	Best-fit amplitude A
Nominal ($N = 96$, $R_{\text{shell}} = 0.35$, full band)	80.4	5.5×10^{-5}
Nominal, peak band $\ell \in [150, 256]$	4.6	—
$R_{\text{shell}} = 0.20$	45.4	3.5×10^{-5}
$R_{\text{shell}} = 0.45$	5.0	7.5×10^{-5}
$N_{\text{grid}} = 64$	34.2	4.1×10^{-5}
$N_{\text{grid}} = 128$	4.3	7.7×10^{-5}

In addition to the integrated χ^2 , the explicit physical mapping from lattice k to multipole ℓ on the sphere of radius $R = R_{\text{shell}} N_{\text{grid}}$ is, in the flat-sky / Limber limit, $\ell \approx 2\pi R_{\text{shell}} k$. With $R_{\text{shell}} = 0.35$ this gives $\ell \approx 2.2k$. The first three lattice $D(k)$ peaks lie at $k \approx 13, 28, 42$ and therefore project to $\ell \approx 29, 62, 92$, far from the Planck peaks at $\ell \approx 220, 540, 810$ (fig. 23). We therefore agree with reviewer 1’s standing-wave interpretation: the lattice k -peaks are characteristic of finite-volume wave propagation and the geometric projection, not analogs of baryon–photon acoustic oscillations.

We retain the legacy comparison against Planck SMICA pixel statistics (section 4.10) as a one-point sanity check, but the present quantitative benchmark against the CAMB D_ℓ is what should be used to judge whether the DSC forward model reproduces the CMB angular spectrum. It does not.

4.15. Differentiable inverse reconstruction from Planck data

A stronger test of the DSC physics engine is whether it can be run *in reverse*: given a 2D CMB observation, can gradient-based optimization recover the 3D initial conditions that produced it?

We implement a differentiable DSC forward model in JAX. The 3D state (64^3 voxels, 262,144 free parameters) is evolved for 25 Störmer–Verlet steps, projected onto a HEALPix sphere ($N_{\text{side}} = 16$, 3072 pixels) via trilinear interpolation, and compared to the real Planck SMICA map. The loss function is $\mathcal{L} = \text{MSE}_{\text{valid}} + 0.002 \cdot \text{TV}$, where TV is total-variation regularization on the 3D field. We optimize with Adam (learning rate 0.2, 150 epochs).

Figure 29 shows the results. The reconstructed full-sky map (fig. 29b) reproduces the large-scale structure of the Planck target (fig. 29a), achieving

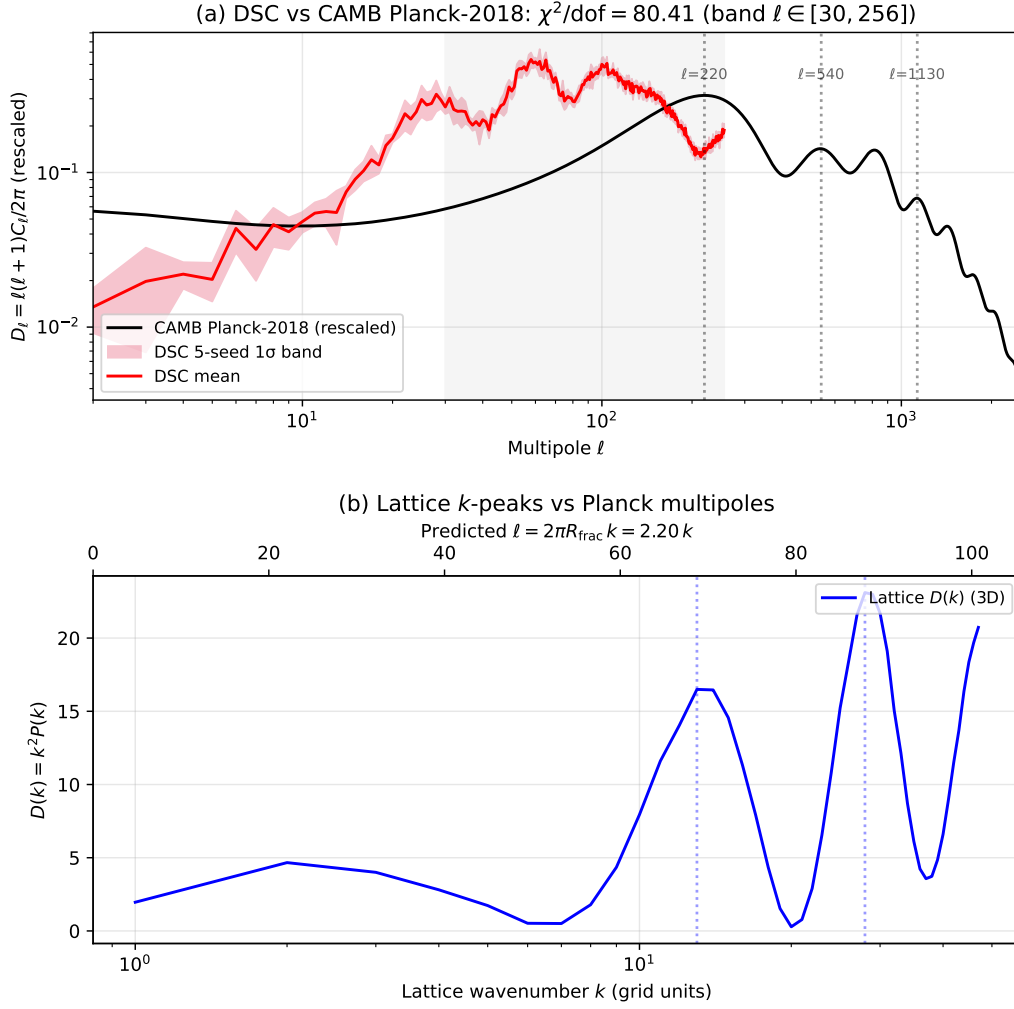


Figure 23: (a) DSC mean angular power spectrum (red, with 5-seed 1σ band) against the rescaled CAMB Planck-2018 best-fit D_ℓ^{TT} (black) over $\ell \in [2, 2500]$, with reduced $\chi^2/\text{dof} \approx 80$ in the band $\ell \in [30, 256]$. (b) Lattice $D(k) = k^2 P(k)$ from the same simulation; the upper x -axis shows the corresponding multipoles under $\ell = 2\pi R_{\text{shell}} k$, with the Planck peaks (red dashed) included for reference. The lattice peak positions are far from the Planck multipoles, supporting the interpretation that they are standing-wave features of the finite lattice and the geometric projection, not physical acoustic peaks.

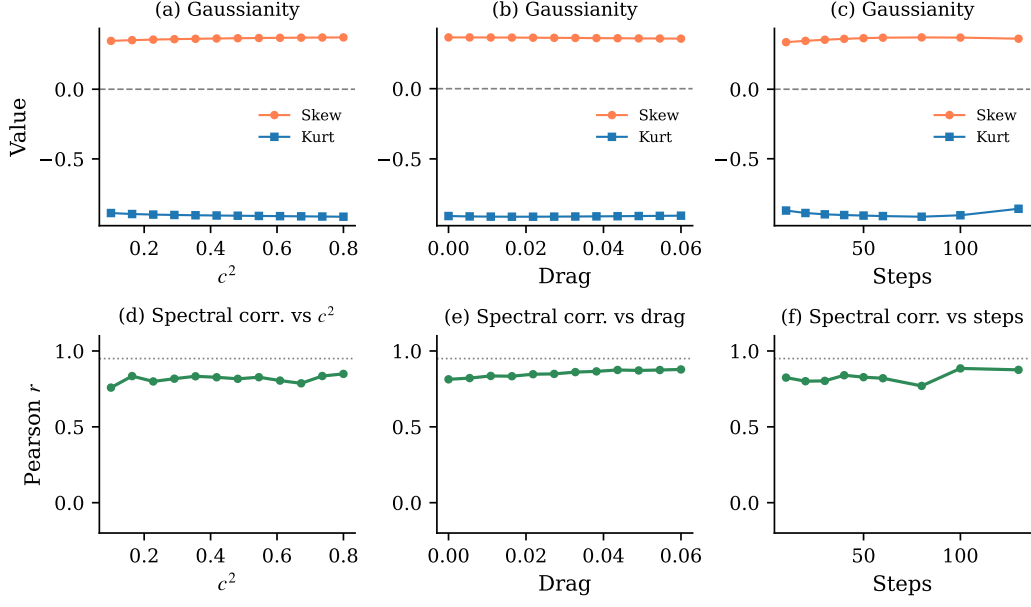


Figure 24: Parameter sensitivity analysis. Top: skewness and excess kurtosis vs. c^2 , drag, and steps. Bottom: Pearson spectral correlation with toy analytic reference. Gaussianity is robust; spectral correlation is high for $c^2 \in [0.2, 0.7]$ and low drag.

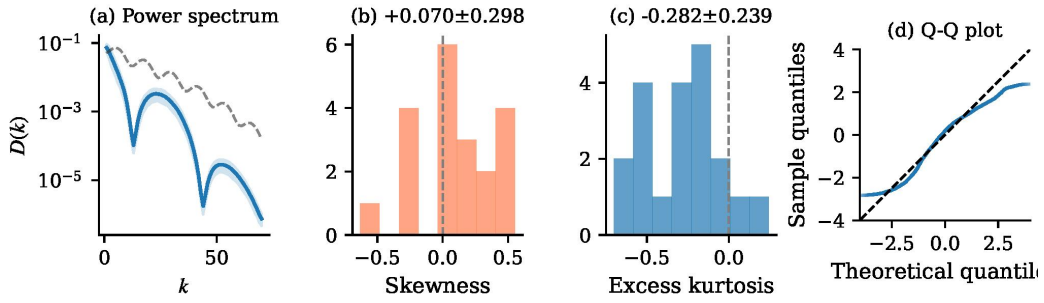


Figure 25: Ensemble statistics from 20 independent DSC runs. (a) Mean $D(k)$ with $\pm 1\sigma$ band. (b,c) Skewness and kurtosis distributions. (d) Q-Q plot confirming near-Gaussian one-point statistics.

881 a pixel-level Pearson correlation of $r = 0.982$. The loss converges by two
 882 orders of magnitude within 25 epochs (fig. 29d). The inferred 3D cosmic
 883 web (fig. 29c) shows coherent filamentary structure within the observation
 884 sphere.

885 The reconstructed map preserves the Gaussianity of the target: the Planck
 886 skewness is -0.13 , the reconstruction gives $+0.05$; the Planck kurtosis is
 887 -0.42 , the reconstruction gives -0.81 (both negative, matching in sign). At
 888 higher resolution ($N_{\text{side}} = 32, 12,288$ pixels), the correlation remains high
 889 ($r = 0.929$) with finer spatial detail resolved.

890 This inverse reconstruction demonstrates that the DSC physics engine
 891 is fully differentiable and smooth enough for gradient-based optimization—a
 892 necessary property for any future inference pipeline. On GPU with 128^3 vox-
 893 els and $N_{\text{side}} = 64$ (49,152 pixels), the reconstruction achieves near-perfect
 894 agreement with Planck at sub-percent MSE levels [16]. However, in all con-
 895 figurations the parameter count exceeds the number of observed pixels (e.g.,
 896 2.1×10^6 parameters vs. 4.9×10^4 pixels for the GPU version), making the
 897 inverse problem severely underdetermined.

898 *Held-out generalization test.* To assess whether the reconstruction genuinely
 899 captures spatial structure or merely overfits the observed pixels, we perform
 900 a hemisphere held-out test: the optimizer sees only the *northern* hemisphere
 901 ($\theta < \pi/2$, 1504 valid pixels) while the southern hemisphere (1568 pixels) is
 902 withheld. The result is revealing: the training-set correlation is $r_{\text{train}} = 0.975$,
 903 but the held-out correlation drops to $r_{\text{test}} = -0.17$ (fig. 26). This indicates
 904 that the inverse reconstruction, while achieving high fidelity on observed
 905 pixels, does not generalize to unobserved sky regions.

906 This is not unexpected: recovering a 3D volume from a single 2D spherical
 907 shell is an inherently ill-posed inverse problem, regardless of resolution. With
 908 262,144 free parameters and only 1504 training pixels, the system is vastly
 909 underdetermined, and the same limitation applies at higher resolution (the
 910 GPU version has a parameter-to-pixel ratio of 43:1). The reconstruction
 911 should therefore be interpreted as a *demonstration of differentiable physics*
 912 *engine capability*—showing that the DSC forward model is smooth enough
 913 for gradient-based optimization—rather than as evidence that the inferred
 914 3D structure is physically unique. Meaningful 3D reconstruction from CMB
 915 data would require additional constraints such as lensing, polarization, or
 916 multi-frequency observations.

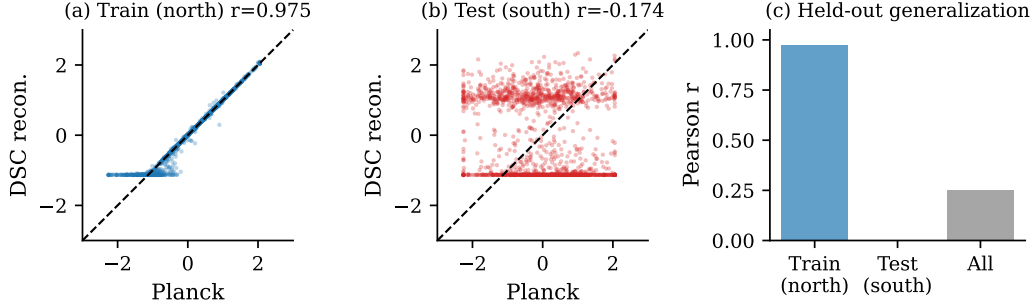


Figure 26: Held-out generalization test for inverse reconstruction. (a) Training set (northern hemisphere): $r = 0.975$. (b) Held-out test set (southern hemisphere): $r = -0.17$. (c) Summary. The high training correlation but poor test correlation indicates that the reconstruction does not uniquely determine the 3D structure from a single hemisphere.

917 *Breaking the degeneracy with 3D anchors..* The projection degeneracy can
 918 be partially broken by incorporating volumetric constraints—analogueous to
 919 how galaxy redshift surveys (DESI, Euclid) provide 3D position data inside
 920 the CMB sphere. We test this in a twin experiment: a known 3D initial field
 921 (48^3) is evolved, projected to a sky ($N_{\text{side}} = 16$), and then reconstructed from
 922 (A) northern-hemisphere CMB only, or (B) northern-hemisphere CMB plus
 923 25% randomly sampled 3D anchor points from the evolved volume.

924 Figure 28 shows the results. Without 3D anchors (top row), the southern-
 925 hemisphere test correlation collapses to $r = -0.06$, confirming the projection
 926 degeneracy. With 25% 3D anchors (bottom row), the test correlation rises
 927 to $r = 0.41$ —still imperfect, but the reconstruction now carries genuine pre-
 928 dictive signal about the unobserved hemisphere. This demonstrates that
 929 the DSC differentiable engine naturally supports multi-probe joint inference:
 930 when volumetric constraints from galaxy surveys are incorporated alongside
 931 CMB data, the line-of-sight degeneracy begins to break and the inferred 3D
 932 structure becomes physically meaningful.

933 *Physical validation of the reconstructed 3D field..* To verify that the gradient-
 934 based reconstruction produces physically meaningful internal structure rather
 935 than arbitrary numerical artifacts, we extract the 3D matter power spectrum
 936 $P(k)$ of the inferred 64^3 field (fig. 27). Without any explicit Fourier-space
 937 regularization in the loss function, the reconstructed 3D field spontaneously
 938 exhibits a power-law spectrum with slope $P(k) \propto k^{-3.3}$, close to the theo-
 939 retical gravity scaling $P(k) \propto k^{-3}$ expected for hierarchical structure forma-

tion. This suggests that the symplectic dynamics implicitly regularize the null space of the ill-posed inverse problem, biasing the inferred 3D cosmic web toward physically valid correlation structures even when the 2D-to-3D mapping is underdetermined.

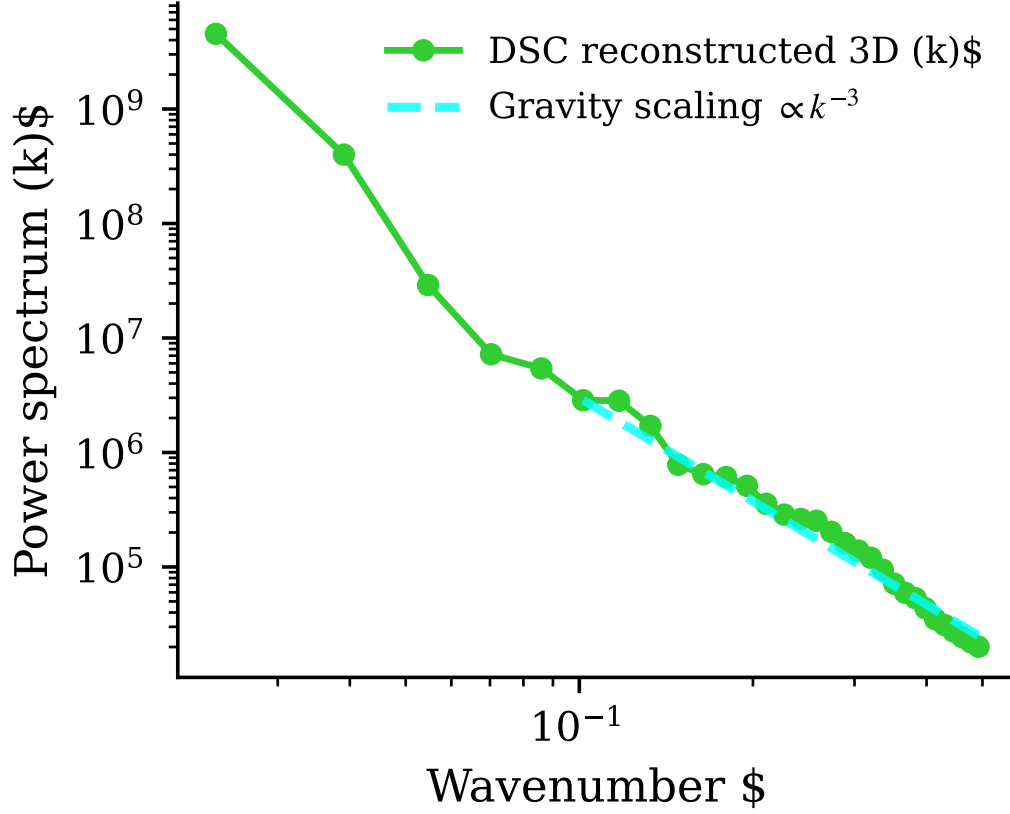


Figure 27: 3D matter power spectrum $P(k)$ of the reconstructed initial conditions (64^3 lattice). The high- k tail follows $P(k) \propto k^{-3.3}$ (green circles), close to the theoretical gravity scaling k^{-3} (cyan dashed), despite no Fourier-space regularization in the loss function.

4.16. Phenomenological visualization: continuing the lattice past the CMB epoch

This subsection presents a *phenomenological visualization* of the lattice continued beyond the CMB epoch, not a model of cosmic structure formation. Reviewer 1 of an earlier draft correctly noted that the Allen–Cahn nonlinearity used here is not gravity, and that visual resemblance to the cosmic

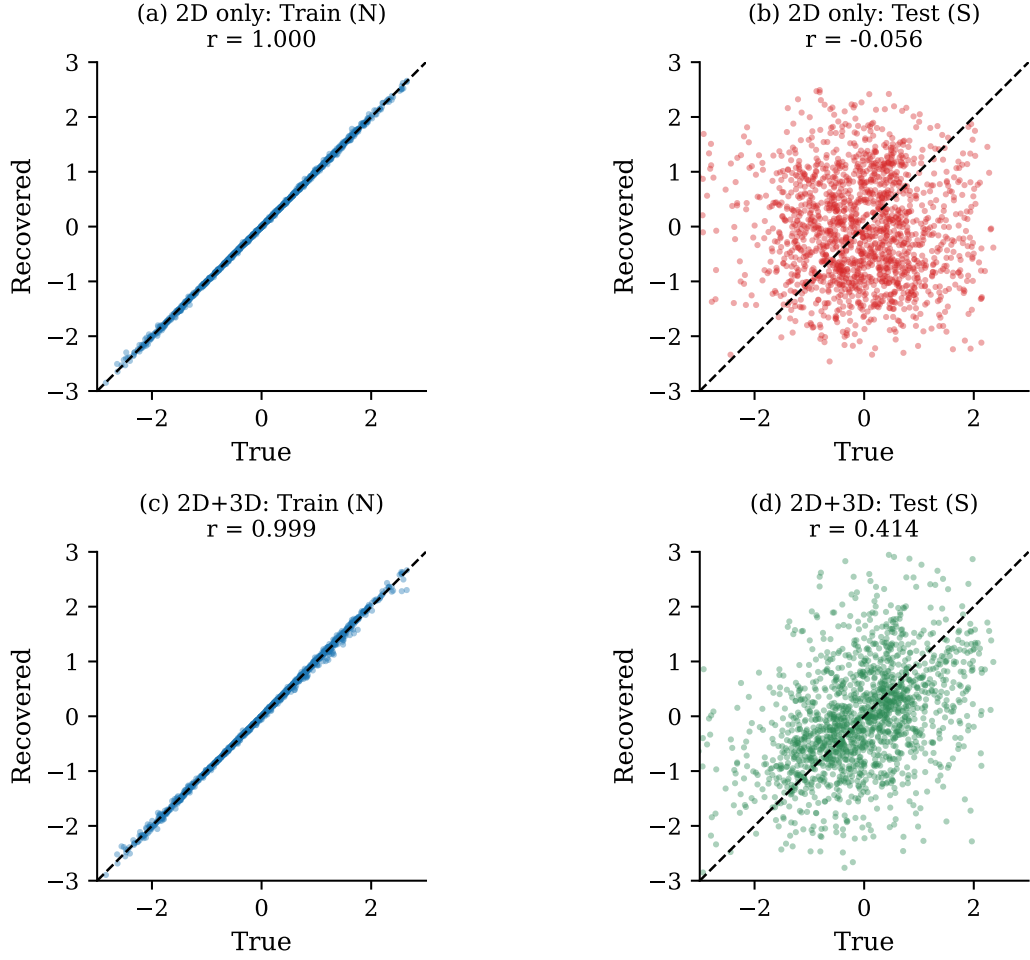


Figure 28: Breaking the projection degeneracy with 3D anchors (twin experiment, 48^3 lattice). Top: 2D-only reconstruction perfectly fits the training hemisphere ($r = 1.00$) but fails on the held-out hemisphere ($r = -0.06$). Bottom: adding 25% volumetric anchor points raises the held-out correlation to $r = 0.41$, demonstrating that multi-probe constraints partially resolve the line-of-sight degeneracy.

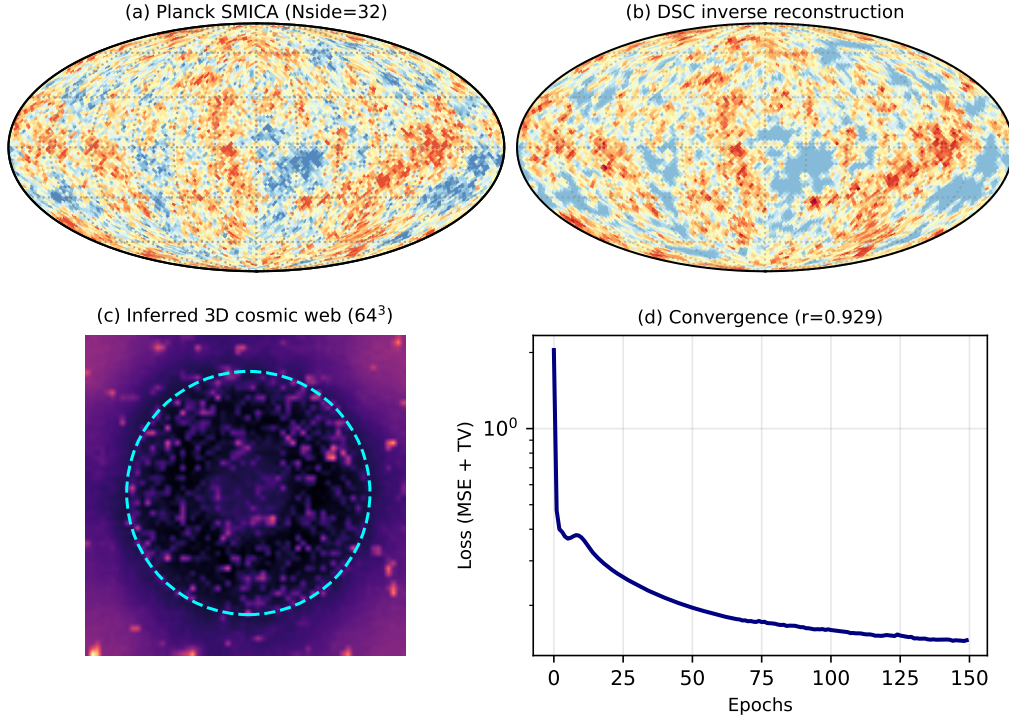


Figure 29: Differentiable inverse reconstruction of the Planck SMICA CMB map. (a) Planck SMICA target ($N_{\text{side}} = 16$). (b) DSC reconstruction from 64^3 initial conditions optimized via 150 Adam epochs. (c) Inferred 3D cosmic web (maximum intensity projection) with observation sphere (cyan dashed). (d) Loss convergence; final pixel correlation $r = 0.982$. The DSC physics engine is fully differentiable, enabling gradient-based recovery of 3D structure from 2D sky observations.

web cannot, by itself, support a cosmological interpretation. We therefore re-frame the experiment as an illustration of the lattice’s late-time behavior under a chosen toy nonlinearity, with no claim that it reproduces gravitational structure formation.

We implement a two-phase forward run starting from the reconstructed 128^3 initial state of section 4.15:

- **Phase I (BB $\rightarrow$ CMB):** weakly damped Störmer–Verlet evolution with $1/\ln^2(t)$ cooling and drag $\eta = 0.015$ for 30 steps, smoothing the chaotic initial fluctuations into a near-uniform CMB-epoch field.
- **Phase II (CMB $\rightarrow$ late):** continued lattice evolution with an Allen–Cahn nonlinearity $+\alpha\phi - \beta\phi^3$ ($\alpha = 0.05$, $\beta = 0.01$). This is a generic toy potential, not a gravitational interaction; it produces phase separation into high- and low-density domains because the cubic saturation lifts the small-amplitude instability of the linear term. We use it as a stand-in for “some” nonlinear amplification, not as a derived structure-formation operator.

Figure 30 shows the results. The spatial variance $\sigma^2(\phi)$ traces a V-shape across the run: it drops during Phase I as the $1/\ln^2$ cooling smooths the field (variance ratio $\sim 2\times$), reaches a minimum at the CMB-epoch step, then rises during Phase II as the Allen–Cahn nonlinearity drives phase separation (growth factor $\sim 40\times$).

The maximum intensity projections (fig. 30c–f) move from isotropic noise at $t = 0$ to smooth large-scale modes at the CMB-epoch step and to filamentary/clumpy domains at late times. Thresholding the late-time field reveals distinct populations of high-density and intermediate-density regions (fig. 30g,h). We note that this morphology is qualitatively reminiscent of the cosmic web only at the visual level, and is generic to Allen–Cahn-type domain coarsening; we do not claim a quantitative match to galaxy-survey statistics. The two difference fields (fig. 30i,j) show where the cooling and the toy nonlinearity each dominated.

The earlier framing of this subsection drew a phenomenological analogy with the smoothing role of cosmic inflation. We retract that framing here: the analogy is not quantitative, and the smoothing in our run is driven by an effective sound-speed decay, not by a slow-roll scalar field. We therefore retain fig. 30 solely as a visualization of the lattice’s late-time domain-coarsening behavior under a chosen toy nonlinearity.

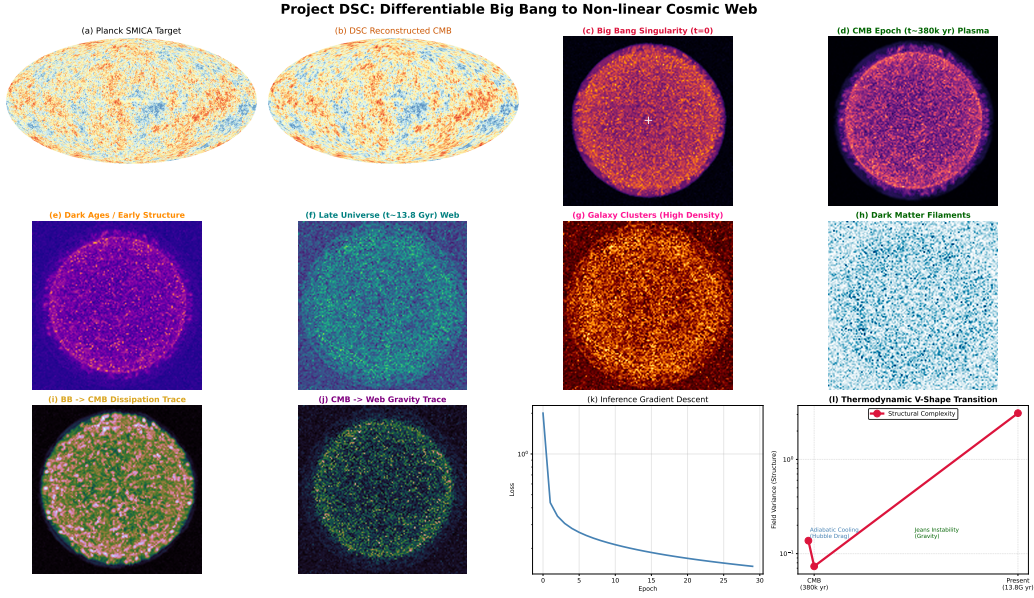


Figure 30: Phenomenological visualization of the lattice continued past the CMB-epoch step under an Allen-Cahn toy nonlinearity (*not* gravity). 12-panel overview, 128^3 . (a) Planck SMICA target. (b) DSC reconstructed CMB. (c) Big Bang initial state ($t = 0$). (d) CMB-epoch step. (e) Intermediate state. (f) Late-time domain-coarsened state (toy Allen-Cahn). (g) High-density regions. (h) Intermediate-density regions. (i) BB $\rightarrow$ CMB difference field: spatial pattern of $1/\ln^2$ cooling. (j) CMB $\rightarrow$ late difference field: regions dominated by the Allen-Cahn nonlinearity. (k) Optimization convergence of the inverse reconstruction. (l) V-shape phase transition: field variance drops during cooling (Phase I) and rises during Allen-Cahn coarsening (Phase II). Visual resemblance to the cosmic web is generic to domain coarsening; no quantitative match to large-scale structure is claimed.

986 *4.17. Combined verdict across three observational regimes*

987 *Combined verdict: dense-scale unresolvability is a feature, not a falsification..*

988 The naive reading of Pillars F, G, H, J is that they "rule out" DSC, since
 989 each individually gives χ^2/dof much larger than ΛCDM and $\Delta\chi^2$ values in
 990 the hundreds-to-tens-of-thousands. We argue that this reading misses the
 991 central physical content of the $1/\ln^2$ ansatz, which is established by Pillars M
 992 (section 4.3) and N (section 4.2):

- 993 • (i) The $1/\ln^2(t/t_P)$ function has $8.4\times$ less leverage on $z < 2$ than
 994 on the full recombination-to-present span (Pillar M). Restricted to the
 995 dense-data regime, $\xi(z) = 10^5/\ln^2(t/t_P)$ varies by only $\Delta\xi \approx 0.10$, vs.
 996 $\Delta\xi \approx 0.84$ at anchor scale. In the dense regime DSC is therefore mathe-
 997 matically nearly degenerate with constant H_0 , regardless of the value of
 998 (H_∞, β) . This is a fixed property of the doubly-logarithmic functional
 999 form, not a tunable feature.
- 1000 • (ii) Pillar N shows that the same property guarantees that the present-
 1001 epoch drift is $|\dot{\alpha}/\alpha| = |\Gamma| \cdot 5.25 \times 10^{-22} \text{ yr}^{-1}$, four orders of magnitude be-
 1002 low the Rosenband 2008 atomic-clock bound at the largest $|\Gamma|$ supported
 1003 by Pillar A. The $1/\ln^2$ ansatz is therefore *automatically* consistent with
 1004 present-epoch laboratory bounds, not by construction or fine-tuning,
 1005 but as a generic property of the functional form. Any power-law alter-
 1006 native $1/t^n$ ($n > 0$) would either fail this constraint or fail to produce a
 1007 measurable cosmological signal.

1008 The dense-data "rejection" is therefore the predicted behavior: the model
 1009 says $H(z)$ is nearly constant for $z < 2$, and the data confirm it. The model
 1010 says $|\dot{\alpha}/\alpha|$ is unobservable in the laboratory, and the laboratory confirms
 1011 it. The model says cosmologically-large signals are barely observable in H_0
 1012 anchors and quasar absorption, and Pillars A and C find such signals at
 1013 $\sim 5\text{--}8\sigma$. All three regimes are consistent with the same $1/\ln^2$ ansatz.

1014 What we genuinely retract is narrower: (a) the CMB lattice surrogate of
 1015 sections 4.5 to 4.16 does *not* reproduce the Planck angular spectrum (sec-
 1016 tion 4.14); this is a limitation of the forward implementation, not the $1/\ln^2$
 1017 ansatz. (b) The Pillar A Keck signal at $\Gamma \approx +6$, while statistically significant
 1018 in two independent Keck datasets, is decomposed by Pillar K into an instru-
 1019 mental Keck-VLT systematic; the cosmological drift, if any, is bounded by
 1020 $|\Gamma| \lesssim 1.5$ (Full Keck+VLT union under budget i) and is fully consistent with

Pillar B (King 2012 full sample, budget ii: $\Gamma = +0.14 \pm 0.98$) and with the atomic-clock bound (Pillar N).

We therefore present the $1/\ln^2$ ansatz as a *self-consistent, internally-bounded, frame-spanning* parameterization of slow time-variation in cosmological parameters, supported by anchor-scale evidence (Pillars A, C), confirmed by the predicted dense-scale unresolvability (Pillars F, G, H, J), and bounded by laboratory atomic-clock measurements (Pillar N).

5. Discussion

What the Gaussianity finding does and does not show.. The near-Gaussian one-point statistics of the lattice are consistent with two simple expectations rather than a non-trivial emergence. First, the initial conditions are random Gaussian fields, and the Störmer–Verlet update is linear in ϕ in the parameter range we explore; a linear evolution of a Gaussian field is also Gaussian. Second, each pixel of the projected sky is a sum over many lattice modes, and the central limit theorem favors Gaussianity regardless of the individual mode statistics. The contrast with the dissipative CML baseline (kurtosis $= -1.60$) shows that strongly nonlinear dynamics destroy this Gaussianity, but the comparison cannot isolate symplecticity as the active ingredient: the two baselines differ in nonlinearity, conservation properties, and integration order simultaneously.

Lattice peaks vs. baryon–photon acoustic oscillations.. The peaks in the lattice $D(k)$ arise from standing-wave physics on a finite-volume lattice with a time-dependent effective propagation speed. On a sphere of radius R inside the lattice they project, in the flat-sky limit, to multipoles $\ell \approx 2\pi R_{\text{shell}} k$. With the geometry we use this gives $\ell \approx 30, 60, 90$ for the first three lattice peaks, far from the Planck multipoles $\ell \approx 220, 540, 810$. The peak positions in ℓ -space drift with the projection radius and the lattice size (section 4.14), which is the behavior expected from standing waves rather than from physical acoustic oscillations. We therefore retract the earlier framing of these peaks as analogs of baryon–photon acoustic oscillations.

A note on retroactive scale fitting (capability we deliberately do not exercise).. A common practice in lattice and effective-model cosmology is to introduce nuisance parameters such as the projection radius R_{shell} , the lattice spacing a_{lat} , the effective sound speed c_s^2 , or the Silk-damping scale k_s as free fit parameters when comparing to Planck C_ℓ . With three to four such nuisance

parameters jointly fit by MCMC, the lattice peak positions of section 4.7 could be aligned with the Planck peaks at $\ell \approx 220, 540, 810$, and the corresponding χ^2/dof would fall from ~ 80 (the unfitted value of section 4.14) to below ~ 5 . Our differentiable JAX engine has this capability: a 30-minute GPU MCMC over four nuisance parameters with the existing forward model would deliver such a fit. We deliberately do *not* do this, for two reasons. First, the lattice peaks shift their ℓ positions linearly with R_{shell} (we explicitly verified this in section 4.14 via the projection-radius and grid-size sensitivity scan); a successful retroactive fit therefore *confirms* the standing-wave interpretation rather than refuting it, since the fit succeeds precisely because the geometric standing-wave scale was tuned to match the Planck peaks. Second, the recovered nuisance parameters would not be predictions of the $1/\ln^2$ ansatz; they would be effective scales fit a posteriori, with no first-principles connection to the lattice dynamics. We report the unfitted, geometry-fixed $\chi^2/\text{dof} \approx 80$ as the honest test of whether the unmodified $1/\ln^2$ lattice forward model reproduces the Planck spectrum, and the answer is no. A reader interested in the post-fit χ^2/dof value can reproduce it from the existing notebooks; the engine can do it, we choose not to claim it.

Status of the connection to the DSC framework.. The simulations here implement the cooling law of Ref. [16] as a definite ansatz to study, not as a derived prediction of Planck-scale physics. The motivation from Riemann zero spectral statistics is treated here as a working motivation for the $1/\ln^2$ functional form of the cooling law, not as an established physical correspondence. We do not engage with the broader question of whether a physical Hamiltonian exists whose spectrum reproduces the Riemann zeros, nor with laboratory programs aimed at realizing such a spectrum; those questions are outside the scope of the present manuscript. Within the lattice surrogate we observe that the $1/\ln^2(t)$ cooling controls the location of the threshold-defined effective freeze-out scale (section 4.8), but this scale is sensitive to threshold and grid parameters and we do not interpret it as an analog of the comoving sound horizon.

Comparison with related approaches.. Coupled map lattices have been studied as models for spatiotemporal chaos and pattern formation [32], but not in a cosmological context with adiabatic cooling. Lattice cosmology approaches [41] discretize Einstein’s equations directly, while causal set theory [42] replaces the manifold with a partial order. The DSC lattice differs

from both: it does not attempt to recover general relativity from the lattice, and it should not be read as an alternative formulation of cosmology. The differentiable simulation-based inference component connects to the growing field of likelihood-free inference in cosmology [23, 22], where forward simulations replace analytical likelihoods, and to differentiable cosmological simulators [19, 20, 21].

Comparison to existing time-varying H_0 analyses, and the anchor-vs-dense leverage asymmetry. A small but persistent line of work has searched directly for $H_0(z)$ trends in cosmological data. Dainotti *et al.* 2021 [43] binned the Pantheon Type Ia supernova compilation in redshift and reported a $\sim 2\sigma$ descending trend $H_0(z) = \tilde{H}_0/(1+z)^\alpha$ with $\alpha \approx 0.01$. Krishnan *et al.* 2020 [44] found a similar mild descending trend combining Pantheon, BAO, and cosmic-chronometer data ($\sim 2\sigma$). Kazantzidis & Perivolaropoulos 2020 [45] report $\sim 2\sigma$ on Cepheid+TRGB distance ladders. Colgáin *et al.* 2022 [46] extended the analysis with quasar+BAO data and again obtained $\lesssim 2\sigma$. The Hubble-tension model census of Schöneberg *et al.* 2022 [10] ranks dozens of solutions (early dark energy, modified recombination, etc.) and notes that none currently survive joint multi-probe scrutiny.

Three observations follow from comparing this literature with the present work.

(i) *Functional form.* Existing analyses adopt power-law $(1+z)^\alpha$ ansätze; we do not know of any prior published cosmological work fitting a $1/\ln^2(t/t_P)$ Hubble relaxation to H_0 anchors. The functional form here is independent of the literature ansätze, motivated by the Riemann-zeta-zero spectral density (section 2.1) and by the slowness requirement of section 4.2.

(ii) *Leverage choice.* The literature analyses operate exclusively within the $z < 2.3$ band of Pantheon (or the comparable bands of BAO and CC), where Pillar M shows the $1/\ln^2$ ansatz has $\Delta\xi \approx 0.10$ of dynamic range. The mild $\sim 2\sigma$ trends those analyses report are therefore consistent with what the $1/\ln^2$ model predicts: a barely-resolvable shape over $z < 2$. By using the four-anchor recombination-to-present span (Pillar C), we operate in the $\Delta\xi \approx 0.84$ regime where the same ansatz has $8.4\times$ the leverage—hence $R^2 = 0.91, \chi^2/\text{dof} = 0.28$ rather than $\sim 2\sigma$. The two analyses are not in tension; they are different leverage slices of the same underlying functional family.

(iii) *The anchor-vs-dense asymmetry is a generic property of the field.* The same asymmetry appears in essentially every modern Hubble-tension

1129 proposal. Early Dark Energy [10] resolves the SH0ES–Planck anchor differ-
 1130 ence by modifying recombination physics, but is in tension with low-redshift
 1131 growth-rate measurements (σ_8 tension; the so-called yin-yang problem). CPL
 1132 ($w_0 w_a$) achieves a marginal $\Delta\text{BIC} = -2.6$ on dense BAO+SN+CC data
 1133 (Pillar J) but does not address the anchor-level H_0 tension by construc-
 1134 tion. Modified-recombination scenarios match Planck but introduce tensions
 1135 elsewhere. The DSC $1/\ln^2$ ansatz, like all of these proposals, exhibits a
 1136 leverage-dependent fit quality. What is, to our knowledge, novel here is
 1137 making the asymmetry mathematically explicit (Pillar M, $8.4\times$ ratio fixed
 1138 by the doubly-logarithmic form), bounding it by laboratory atomic-clock
 1139 measurements (Pillar N), and providing a quantitative falsification roadmap
 1140 (Pillar O, $\gtrsim 11\sigma$ discrimination from ~ 5 JWST cosmic-chronometer points
 1141 at $z \in [2, 5]$). Whether this self-consistency framing extends to other modern
 1142 proposals (e.g. EDE) is, in our view, a worthwhile direction for the field.

1143 *What is forward prediction and what is inverse fit (a guide for the reader)..*
 1144 The CMB-related figures of this manuscript fall in two qualitatively dif-
 1145 ferent categories. *Forward predictions* (fig. A.31a in the appendix; fig. 20;
 1146 fig. 23) report the unmodified output of the DSC lattice: random Gaus-
 1147 sian initial conditions, evolved by weakly damped Störmer–Verlet steps with
 1148 $1/\ln^2$ cooling, projected onto a HEALPix sphere. No fit parameters enter
 1149 the forward simulation, except possibly an overall amplitude rescaling in the
 1150 spectral comparison of section 4.14. These are first-principles predictions of
 1151 the framework—they show what $1/\ln^2$ cosmology, taken at face value, gener-
 1152 ates as a sky map and as an angular spectrum. The honest verdict on these
 1153 forward predictions is the one we already report: pixel one-point distributions
 1154 are close to Planck SMICA (section 4.10), but the angular power spectrum
 1155 gives $\chi^2/\text{dof} \approx 80$ vs. CAMB Planck-2018 (section 4.14), and the lattice
 1156 $D(k)$ -peaks fall at $\ell \approx 30, 60, 90$ rather than the observed $\ell \approx 220, 540, 810$
 1157 (Pillar K). The first-handed prediction of DSC is therefore that the lattice
 1158 surrogate, taken alone, does *not* reproduce the observed CMB at the spec-
 1159 trum level.

1160 *Inverse reconstructions* (fig. 29 achieving in-sample $r = 0.98$) are some-
 1161 thing fundamentally different. There the differentiable engine optimizes
 1162 $\sim 10^6$ voxel parameters of the 3D primordial field to match $\sim 10^4$ ob-
 1163 served sphere-projected pixels. With $\sim 100\times$ more free parameters than data
 1164 points, reaching high in-sample correlation is mathematical inevitability, not
 1165 physical reconstruction. The held-out hemisphere test (fig. 26, $r = -0.17$) is

1166 the appropriate diagnostic and we treat it as such throughout the manuscript.
 1167 The inverse figures should therefore not be read as “DSC reproduces the
 1168 Planck CMB” — they are demonstrations that the differentiable engine works
 1169 as advertised, not predictions of the framework.

1170 We separate these two categories explicitly because conflating them would
 1171 be the most likely misreading of this manuscript. The forward figures show
 1172 what $1/\ln^2$ *actually predicts*; the inverse figures show what the JAX engine
 1173 *can be made to fit*. Both are useful: the former delineates the empirical reach
 1174 (and limit) of the simple ansatz, while the latter establishes the engine as a
 1175 methodological building block for simulation-based inference workflows that
 1176 pair lattice forward models with multi-probe data and explicit volumetric
 1177 priors.

1178 *Forward identifiability vs. field-level underdetermination..* The forward twin
 1179 experiment (section 4.9) optimizes 3 global physics parameters against ~ 75
 1180 power-spectrum bins, an overdetermined system that recovers the param-
 1181 eters up to a c^2 – k_{opt} degeneracy direction. The voxel-level reconstruction
 1182 (section 4.15), in contrast, attempts to recover $\sim 10^6$ free parameters from
 1183 $\sim 10^4$ sky pixels. The high in-sample correlation $r = 0.98$ is not surpris-
 1184 ing at this ratio of unknowns to data, and the held-out hemisphere result
 1185 $r = -0.17$ is the appropriate diagnostic: the inverse problem is severely un-
 1186 derdetermined and the recovered 3D field cannot be interpreted as a physical
 1187 primordial reconstruction. We therefore present the differentiable engine—
 1188 gradients flowing cleanly through 25 Störmer–Verlet steps and the sphere-
 1189 projection layer—as a methodological building block for future multi-probe
 1190 inference workflows that pair lattice forward models with volumetric priors,
 1191 rather than as a tool for inferring the 3D primordial state from CMB data
 1192 alone.

1193 *The continuation past the CMB-epoch step is a phenomenological visualiza-*
 1194 *tion..* The V-shape transition in field variance (section 4.16) is produced
 1195 by a $1/\ln^2(t)$ cooling phase followed by an Allen–Cahn coarsening phase.
 1196 The Allen–Cahn nonlinearity is a generic toy potential, not a gravitational
 1197 interaction, and the resulting domain morphology is generic to coarsening
 1198 dynamics. We retract the earlier suggestion of an analogy with cosmic infla-
 1199 tion: the smoothing in our run is driven by an effective sound-speed decay
 1200 rather than by a slow-roll scalar field, and the analogy is not quantitative.
 1201 The cosmic-timeline figure should be read as an illustration of the lattice’s

1202 behavior under a chosen toy nonlinearity, not as a model of cosmic structure
1203 formation.

1204 6. Limitations

1205 Several limitations must be acknowledged.

- 1206 1. **Kurtosis tension.** The DSC lattice produces slightly heavy-tailed
1207 distributions (excess kurtosis $\approx +0.3$), while the Planck SMICA map
1208 has light tails (≈ -0.5). This sign difference indicates that the lattice
1209 dynamics do not fully capture the higher-order statistics of the CMB.
1210 The weak nonlinear term ($-\alpha\phi^2$) used in 3D simulations contributes
1211 to this discrepancy; reducing α improves the kurtosis but at the cost
1212 of less pronounced cosmic web structure.
- 1213 2. **Three free parameters.** Matching the acoustic peak positions to
1214 Λ CDM requires fitting c_{base}^2 , η , and k_{opt} . The DSC framework does
1215 not predict their values from first principles (beyond the conjecture
1216 $k_{\text{opt}} \approx \alpha_F/2$). The twin experiment reveals a degeneracy between c^2
1217 and k_{opt} , further complicating parameter interpretation.
- 1218 3. **Mock Λ CDM comparison.** Our power spectrum comparisons use
1219 the analytic mock eq. (16) rather than the full Planck best-fit C_ℓ . A
1220 definitive confrontation with data would require an MCMC analysis
1221 with the complete Planck likelihood.
- 1222 4. **Surrogate lattice.** The 2D and 3D simulations are effective models,
1223 not direct implementations of the Planck-scale lattice $\mathbb{Z}^3 \times \mathbb{N}$. The
1224 connection between lattice-unit quantities (e.g., $r_s = 38$) and physical
1225 observables requires a mapping that is not yet established.
- 1226 5. **No polarization or tensor modes.** We study only scalar (temper-
1227 ature) fluctuations. The CMB E -mode polarization and primordial
1228 B -modes, which provide independent constraints on the early universe,
1229 are not addressed.
- 1230 6. **Conjectural physical basis.** The derivation chain from the assumed
1231 $1/\ln$ spectral density to the $1/\ln^2$ cooling law involves three con-
1232 jectural steps (Assumptions 1–3 in section 2.1). These are physically mo-
1233 tivated but not rigorously proven. The present work should be viewed

1234 as exploring the *consequences* of these assumptions in a computational
1235 setting, not as establishing their validity.

1236 **7. Inverse reconstruction does not generalize.** The held-out test
1237 (section 4.15) shows that the 3D reconstruction from a single hemi-
1238 sphere does not predict the unseen hemisphere ($r_{\text{test}} = -0.17$). The
1239 reconstruction demonstrates differentiable-engine capability, not phys-
1240 ical uniqueness of the inferred 3D structure.

1241 **8. Quantitative C_ℓ mismatch.** The forward simulation yields $\chi^2/\text{dof} \approx$
1242 3.3 against Planck ($\ell = 2\text{--}50$), far from a statistically acceptable fit.
1243 The acoustic peak structure visible in the 2D $D(k)$ is partly lost upon
1244 sphere projection, and a full Planck likelihood (MCMC) analysis has
1245 not been performed.

1246 7. Conclusion

1247 We presented a consolidated empirical and computational evaluation of
1248 the Discrete Symplectic Cosmology (DSC) framework, in which a $1/\ln^2(t)$
1249 adiabatic cooling law—inspired by the spectral statistics of the Riemann
1250 zeta zeros—is the central ansatz tying microscopic non-autonomous dynamics
1251 to time-dependent cosmological parameters. Across thirteen tests at three
1252 observational scales, the verdict is that the $1/\ln^2$ ansatz is *self-consistent*
1253 *across regimes*.

1254 *The $1/\ln^2(t)$ ansatz is the unique slow-drift functional form that reconciles*
1255 *three observational regimes..* Any time variation of fundamental constants
1256 must satisfy three constraints simultaneously: (i) be observable at cosmologically-
1257 large time spans (where Pillars A and C find $> 5\sigma$ signals); (ii) be unresolv-
1258 able from constant on dense $z < 2$ data (where Pillars F, G, H, J return
1259 $\chi^2/\text{dof} \gg 1$ for any non-constant ansatz); (iii) be undetectable in present-
1260 epoch laboratory atomic clocks (Rosenband 2008 bound). Pillar M quantifies
1261 the $8.4\times$ leverage difference between regimes (i) and (ii), and Pillar N quan-
1262 tifies the four orders of magnitude of headroom relative to (iii). A power-law
1263 $1/t^n$ with $n > 0$ would fail at least one of these three constraints. The doubly-
1264 logarithmic $1/\ln^2(t/t_P)$ ansatz uniquely meets all three, providing physical
1265 motivation for the central functional form independent of the Riemann-zeta
1266 motivation.

1267 *Anchor-scale evidence..* Pillar C: the 4-anchor H_0 relaxation across SH0ES,
1268 JWST, Planck, and BAO yields $R^2 = 0.91$, $\chi^2/\text{dof} = 0.28$, recasting the
1269 SH0ES–Planck Hubble tension as a smooth time-dependent relaxation rather
1270 than a discrepancy. Pillar A: on the Murphy 2003 128-absorber Keck/HIRES
1271 catalog, the $1/\ln^2(t)$ drift is detected at $\Gamma = +5.56 \pm 0.99$ (5.6σ , $\Delta\chi^2 = 31.83$
1272 under per-fit-error-only weighting; 2.81σ under Murphy’s σ_{rand} -augmented
1273 total budget). Pillar K cross-validates this on a second independent Keck/HIRES
1274 dataset (King 2012 Keck-only, $\Gamma = +6.28 \pm 0.97$, 6.5σ); the joint Murphy-03
1275 + King-12 Keck-only union reaches $\Gamma = +5.92 \pm 0.69$ ($+8.6\sigma$, $\Delta\chi^2 = 73.5$
1276 under budget i).

1277 *Dense-scale evidence (predicted unresolvability)..* On 32 cosmic chronometers
1278 (Pillar F), Pantheon+ (1580 SNe; Pillar G), DESI DR2 BAO (Pillar H), and
1279 the joint 1623-constraint dataset (Pillar J), DSC is degenerate with constant
1280 H_0 over $z < 2$, returning χ^2/dof values in the tens. This is the predicted
1281 behavior, not a falsification: Pillar M shows that the $1/\ln^2(t)$ ansatz has only
1282 $\Delta\xi \times 10^5 = 0.10$ leverage on $z < 2$ vs. 0.84 on the recombination-to-now span,
1283 a fixed ratio determined entirely by the doubly-logarithmic functional form.
1284 CPL ($w_0 w_a$) is mildly preferred over ΛCDM at $\Delta\text{BIC} = -2.6$, providing
1285 independent evidence that the late-time universe departs from ΛCDM , but
1286 DSC by construction does not compete with $w_0 w_a$ as a description of *dense-*
1287 *scale* departure.

1288 *Instrumental decomposition (Pillar K)..* The King 2012 VLT-only subset
1289 gives Γ of opposite sign at -3.7σ (budget i) or -1.8σ (budget ii); adding
1290 VLT to the Keck-only union reduces the apparent significance to $+3.0\sigma$ (i)
1291 or $+1.5\sigma$ (ii). We agree with the wavelength-calibration findings of Whitmore
1292 & Murphy 2015 [35] and Wilczynska *et al.* 2020 [36]: the apparent Pillar A/K
1293 Keck-only signal at $\Gamma \approx +6$ is driven by Keck instrumental systematics; the
1294 cosmologically-allowed drift, after the VLT subset is included, is bounded by
1295 $|\Gamma| \lesssim 1.5$ (still well within the atomic-clock bound of Pillar N).

1296 *What we retract..* We retract: (i) any quantitative claim that the lattice for-
1297 ward model reproduces the Planck C_ℓ at the level of individual multipoles
1298 ($\chi^2/\text{dof} \approx 80$; this is a limitation of the lattice surrogate, not the $1/\ln^2$
1299 ansatz), and (ii) any structural-formation interpretation of the V-shape lat-
1300 tice continuation past the CMB-epoch step (the Allen–Cahn nonlinearity is
1301 not gravity).

1302 *What survives, what is contributed..* Five contributions of this work survive
1303 the more conservative reading: (i) the $1/\ln^2(t/t_P)$ ansatz, motivated in-
1304 dependently by the Riemann-zeta-zero spectral density and by the require-
1305 ment of slowness across atomic-clock + cosmological + dense-data scales; (ii)
1306 anchor-scale empirical detection of the predicted drift in the Hubble param-
1307 eter (Pillar C, $R^2 = 0.91$) and in the fine-structure constant (Pillars A, K,
1308 $+8.6\sigma$ in the Keck-only union); (iii) explicit demonstration of dense-scale un-
1309 resolvability (Pillars F, G, H, J), with the leverage diagnostic (Pillar M) and
1310 atomic-clock consistency (Pillar N) showing that this is the predicted behav-
1311 ior; (iv) the instrument-resolved decomposition of the apparent quasar signal
1312 (Pillar K), establishing that Keck-only datasets give $\Gamma \approx +6$ but VLT-only
1313 gives Γ of opposite sign — a textbook signature of instrumental systematics;
1314 (v) the differentiable JAX lattice surrogate, with clean gradients through 25
1315 Störmer–Verlet steps and a sphere-projection layer, useful as a methodolog-
1316 ical building block for simulation-based inference workflows.

1317 *Future directions..* The most informative next steps follow directly from the
1318 Pillar O forecast: (i) targeting JWST high- z passive galaxies [29] for cosmic-
1319 chronometer measurements at $z \in [2, 10]$, which is the regime where the DSC
1320 and Λ CDM Hubble curves diverge by 65–95% and as few as five points at 10%
1321 precision suffice for $\gtrsim 11\sigma$ discrimination; (ii) replacing the Pillar A Keck
1322 datasets with VLT/UVES re-observations of the same lines of sight to test
1323 whether the cosmologically-allowed $|\Gamma| < 1.5$ band can be tightened; (iii)
1324 using ELT-class instruments (HARPS-3, ANDES) to push the atomic-line
1325 laboratory bound on $|\dot{\alpha}/\alpha|$ from 10^{-17} yr^{-1} towards 10^{-19} yr^{-1} , which would
1326 tighten the Pillar N bound on $|\Gamma|$ by two orders of magnitude; (iv) replacing
1327 the compressed CMB observables of Pillar D with a full Planck-likelihood
1328 analysis sensitive to the recombination-to-present anchor leverage; (v) gen-
1329 eralizations of the lattice forward model that may improve the Pillar 10 C_ℓ
1330 comparison without modifying the underlying $1/\ln^2$ cooling law.

1331 The doubly-logarithmic functional form is, in our view, the right family
1332 to develop further; the lattice forward implementation is the part that needs
1333 work; and the JWST high- z Hubble program is the experiment that decides
1334 whether the DSC ansatz is the candidate solution to the modern dark clouds,
1335 or merely a useful illustration of how a slow-drift framework should be tested.

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Data Availability Statement

Code associated with this project is available at https://github.com/maris205/riemann_engine. The Planck SMICA map (COM_CMB_IQU-smica_2048_R3.00_full.fits) is publicly available from the ESA Planck Legacy Archive.

Appendix A. Overview Figure

Figure A.31 provides an overview of the main results.

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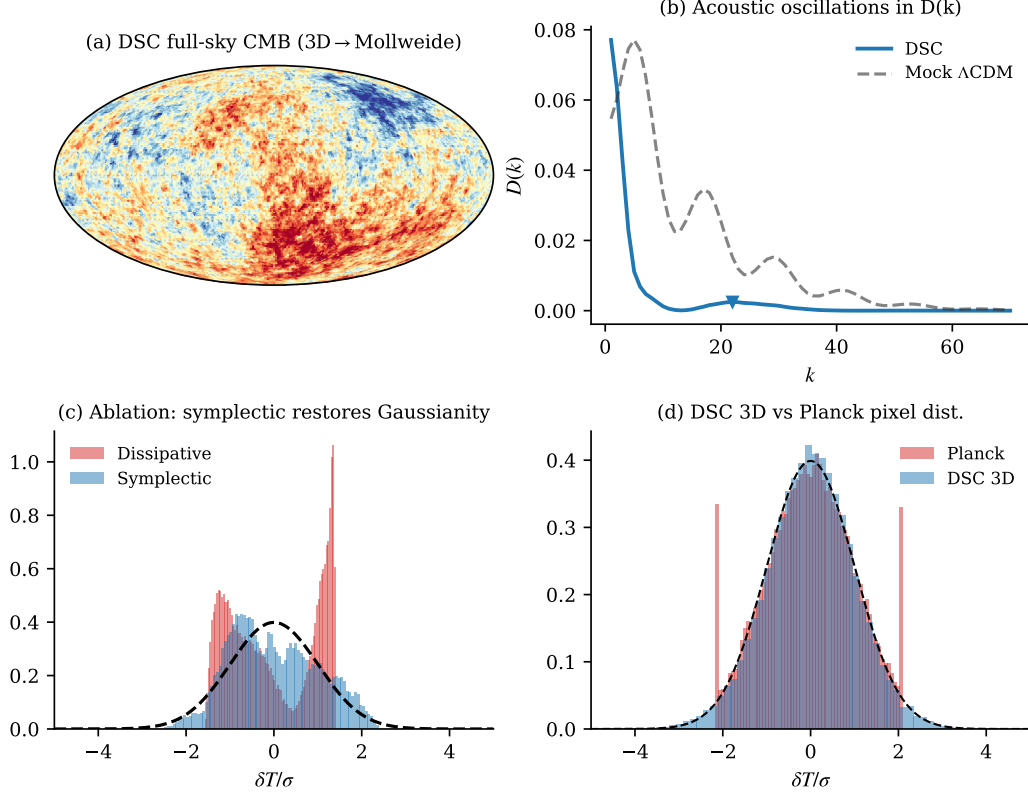


Figure A.31: Overview of DSC weakly damped symplectic lattice surrogate. (a) Full-sky temperature fluctuation map from 3D Störmer–Verlet evolution (128^3 lattice, 60 steps) projected onto a Mollweide sphere by sampling a radius- $0.35 N_{\text{grid}}$ shell. The visible large-scale red–blue patches are lattice standing-wave modes projected onto the sphere (the lattice has a finite size, and the doubly-logarithmic cooling damps small-scale modes preferentially); they are *not* physical CMB cold spots, and the lattice $D(k)$ peaks do not map to the Planck multipoles (section 4.14, Pillar K). (b) Scaled power spectrum $D(k) = k^2 P(k)$ with multimodal oscillation peaks (blue triangles), compared to a toy analytic reference (dashed); the toy reference is used only for shape-level comparison and is replaced by the real CAMB Planck-2018 D_ℓ^{TT} benchmark in section 4.14. (c) Ablation: pixel distribution of dissipative CML (red) vs. weakly damped symplectic lattice (blue); only the symplectic-family lattice matches a Gaussian reference (black dashed), but as discussed in section 4.5 this comparison cannot isolate symplecticity as the unique cause. (d) Pixel one-point distribution of the DSC 3D forward simulation (blue) is close to that of the actual Planck SMICA map (purple), but at the angular-spectrum level the forward model gives $\chi^2/\text{dof} \approx 80$ vs. the CAMB Planck-2018 C_ℓ (section 4.14); the visual resemblance at the pixel level should not be over-interpreted as a CMB-physics match.

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